

Speaker Identity or Communicative Style? Pitch, Prediction, and Speaker Confounds in U.S. Presidential Debates, 1988–2020

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Abstract

Do linguistic and vocal properties distinguish candidates whose tickets went on to win a U.S. presidential election from those whose tickets did not? We study every televised U.S. general-election presidential debate from 1988 to 2020: **24** debates, **29,672** time-aligned sentences, of which **24,193** are spoken by candidates.

We first compare candidate speech with a pooled, uncontrolled text-based profile. Several of its differences shrink, disappear, or reverse once we repeat the same comparison within debates. We then screen **43** linguistic, semantic, discourse and acoustic measures for a within-debate difference between winning and non-winning candidates. One property survives: a specific feature of the pitch contour, which we characterise in detail.

Generalisation to an unseen speaker or debate cannot be assumed; it has to be tested directly. So we evaluate every predictive model under two holdout schemes: holding out a debate, and holding out every debate containing either of its speakers. We apply the same test to a large language model asked to judge the same debates, and we add a contamination probe and a closed-vocabulary grounding check on its stated reasons.

Winning candidates carry less residual pitch movement than non-winning candidates, once each sentence’s own rise or fall is removed. Of **53** properties tested, this is the only one that survives every validity and stability check we run, including dropping any one of the corpus’s fourteen speakers in turn. But it does not predict a debate we have not seen: its **95%** prediction interval includes zero. The reason is structural, not statistical. Speaker identity alone explains **81%** of the outcome label, in a corpus where only fourteen people have ever taken this stage.

The same limitation defeats three classical models: a decision forest, a linear discriminant, and a weighted sum. All three call debates correctly, until every debate containing either candidate’s own voice is removed from training. Then all three fall to chance. A large language model shows the same pattern from a different angle. Given an anonymised transcript, it calls the winner in **24** of **24** debates. But asked instead, with no prediction question attached, to name the debate itself, it also succeeds in **24** of **24** cases. Given only numeric descriptors, with nothing to identify, its accuracy falls back to chance. In this corpus, a model’s apparent success is not evidence of having learned how winners speak; three times over, it is a sign of what the model was allowed to see.

Keywords: Political discourse analysis, Spoken corpus construction, Prosody, Pitch range, Dialogue-act annotation, Multimodal analysis, Speaker confound, Measurement validity

1 Introduction

A broad multimodal screen of U.S. presidential debate speech turns up one coherent, outcome-associated pattern: candidates whose tickets went on to win show less residual, sentence-level pitch movement than candidates whose tickets did not. The pattern survives an unusually strong battery of measurement checks. It does not support reliable prediction for an unseen speaker or an unseen debate. When the same prediction question is handed to classical statistical models and to a large language model, both appear to succeed for the same avoidable reason: speaker or transcript identity leaks into what should have been a held-out fold. This paper is organised around three connected claims: a substantive discovery, a methodological diagnosis of why classical models mislead, and the same diagnosis run again on a language model. This is the order in which the evidence actually has to be read: found, checked, tested for generalisation, and checked once more against the newest kind of instrument available for checking it.

1.1 What Debate Analysis Is

Debate analysis studies what happens when two people argue in public under fixed rules: what was said, how it was said, and what effect it had. Traditionally this was done by hand, with a researcher watching, transcribing and coding. Computational debate analysis does the same work at scale: programs measure properties of the recording and the transcript. This approach can therefore ask questions across decades of material rather than across a single evening.

The move to computation buys coverage and costs certainty. A human coder who marks a sentence as an appeal to patriotism can say why. A program that marks the same sentence has applied a rule. Whether that rule captures the idea is a separate question that usually goes unasked. Much of this paper is about what happens when it is asked.

Televised general-election debates are among the most watched political events in the United States. They reach tens of millions of viewers. They are one of the few occasions on which voters see the candidates side by side. They are credited with shifting voter intention [1].

For a language researcher they offer something more specific: two candidates answer the same questions, in the same room, at the same time, before the same audience, recorded through the same equipment. Topic, setting, interlocutor and recording chain, the things that normally make spoken language hard to compare, are held fixed by the format. We call this a *standardised naturalistic comparison* rather than a controlled experiment. Nobody assigns which candidate stands on which side, and nobody assigns who wins. What varies, within a debate, is the two people.

The U.S. case adds three further properties. Debates have been held on a regular cycle since 1988, so a corpus can span decades with a consistent format. Transcripts and recordings are public. Each debate is followed, within weeks, by an election. The election attaches an objective outcome to each speaker, something almost no other spoken corpus provides.

1.2 Why Language and Voice Must Be Examined Together

A spoken utterance can be described at several levels at once. Each level answers a different question about the speaker.

Linguistic analysis asks how an argument is built. Counting clauses, measuring how deeply they nest, and measuring how far a word sits from the word it depends on are the standard ways to quantify syntactic complexity. The last of these has an established reading in terms of processing effort [2]. These are structural properties that barely depend on topic. That makes them comparable across debates about different things.

Semantic analysis asks what was said. Word lists give interpretable counts of sentiment and topic but detect vocabulary rather than meaning. Sentence embeddings capture meaning across paraphrase but are opaque. Word class and pronoun use indicate whether a speaker is talking about the people in the room or about things and ideas [3].

Acoustic analysis asks how it was said. Pitch and how much it moves describe intonation. Measures of the voice source describe whether a voice sounds clear, breathy or harsh [4]. Silent pauses indicate where planning effort falls [5, 6]. None of this survives transcription, so a text-only study discards it entirely.

A fourth layer, *discourse annotation*, labels what an utterance does: the act it performs and the rhetorical move it makes. This is where political discourse analysis has traditionally worked. Section 5 shows that it is also the layer where the instrument matters more than the effect.

Three lines of prior work make the case for measuring all of this together rather than any one layer alone. Section 2 covers them in detail: political communication research on what debates convey [1], speech science on the prosodic correlates of perceived charisma [7, 8], and computational text analysis on what word-level features carry [3, 9].

1.3 Gaps in Previous Research

Four gaps motivated this study: scale and span, modality separation, speaker–outcome confounding, and instrument validity.

Scale and span. Most multimodal debate studies examine one debate or one election cycle. Nobody has measured every general-election presidential debate over three decades on a common footing.

Modality separation. Acoustic studies rarely measure syntax. Text studies discard the audio. Where a difference is found at one level, there is no way to tell whether it is the same difference showing up twice.

Speaker–outcome confounding. A candidate’s speech and a candidate’s identity arrive together. Studies that compare groups of politicians rarely report how many distinct people underlie a reported difference. A model trained to distinguish the groups may be learning to recognise the politicians rather than a general pattern of political communication.

Instrument validity. Keyword rules, acoustic descriptors and, more recently, language-model judgments are all used as measurements of theoretical constructs. The check that would discipline that use is usually unavailable or simply unasked. Does the measure covary with something external? Does it survive being asked why it answered that way?

1.4 Research Questions

RQ1. Which linguistic, semantic, discourse and acoustic properties differ, within debates, between candidates whose tickets subsequently won and those whose tickets did not?

RQ2. What specific property of the pitch contour accounts for the strongest and most robust acoustic difference?

RQ3. Do the individual and combined measures that appear to distinguish the two groups generalise to debates involving previously unseen speakers?

RQ4. Can a large language model judge electoral outcome from debate material without relying on transcript memorisation or unsupported explanations?

One further question is exploratory rather than primary, because the corpus can only partly answer it: *for the small number of candidates observed in both a winning and a losing election cycle, do their speech characteristics change with the outcome?* Only George H.W. Bush and Donald Trump provide both conditions. Section 7 shows that their results disagree. This question is therefore reported as a case study (Section 7, Appendix H), not as evidence either way.

1.5 Contributions

1. A multimodal, sentence-aligned corpus of all 24 televised U.S. general-election presidential debates from 1988 to 2020, with an objective, externally determined outcome label attached to every candidate—whether their ticket won the election that followed. This label is not a measure of speaking well. No claim in this paper depends on winning and speaking well being the same thing.

Table 1 One failure mode, tested four ways. “Looks like it works” is each method’s own best evaluation. “Real test” removes the information the method was actually using. AUC (area under the curve) is the chance that the model ranks a random winning-candidate sentence above a random non-winning one; 0.5 is chance level, 1.0 is perfect. Full results in Section 7.

Method	Looks like it works	Real test
Weighted sum	17/24 debates, holding out only the debate	11/24 holding out both speakers: chance
Linear discriminant	19/24 debates ($p = 0.0066$)	10/24: chance
Decision forest	17/24 debates; sentence AUC 0.630	12/24; AUC chance
LLM, transcript only	24/24 debates	Asked to name the debate instead of the winner: 24/24 — it already knew

2. A pooled, uncontrolled text-based comparison of candidate speech by outcome (sentence form, syntax, lexical composition and discourse-tag frequency), reported alongside a common within-debate statistical screen of 43 linguistic, semantic, discourse and acoustic measures (RQ1). This lets us judge which differences survive correction for having looked at 43 things against which differences a naive comparison would have reported instead.
3. A targeted characterisation of residual pitch control using ten higher-order and dynamic measures of the same pitch contour (RQ2). This shows that the surviving difference is specifically about the movement left over once each sentence’s own trend is removed, not about speed, smoothness, or overall range.
4. A speaker-aware and contamination-aware evaluation protocol (RQ3, RQ4). It shows how classical predictive models and a large language model can both appear to generalise when they are in fact exploiting speaker or transcript identity, and it presents the cheap checks that catch this in either case.

1.6 Overview of the Argument

Table 1 previews the shape of the result before the paper builds it in full, one section at a time. Four methods, three classical and one a large language model, are each asked the same question: given a debate, which candidate went on to win? All four look like they can answer it. None of them can, once the test is changed to keep either candidate’s own voice, or the model’s memory of the transcript, out of the picture.

The first three rows fail for the same reason: the model was never learning a style of speaking, only which of fourteen people was talking (Section 7). The fourth fails for a related but distinct reason: the transcript is one of the most quoted texts in existence. A model asked to name the winner therefore has an easier, unintended route to the

right answer (Section 7.7). The rest of this paper is the evidence behind this table, built up one result at a time.

2 Related Work

2.1 Computational Analysis of U.S. Political Debates

Benoit [1] establishes the substantive case: debates convey information about policy positions and personal character. Voters use this information. This is a content-level result. It says nothing about delivery, which is why delivery still needs to be measured separately. Two properties of the broader literature on debate effects and political speech shaped our design. First, studies are typically confined to one election or one debate, so nothing is known about whether a difference found in one cycle recurs in another. Second, the number of distinct speakers behind a reported result is rarely stated. In a corpus of political speech, speakers are few and memorable. A difference between two groups of politicians may just be a difference between two sets of individuals.

2.2 Prosody, Charisma, and Political Speech

The acoustic line begins with Rosenberg and Hirschberg [7], who identified prosodic correlates of perceived charisma in political speech: pitch dynamics, intensity range and pause structure. Their feature set is close to ours. Their finding motivated our expectation that pitch and pausing would matter most. Gravano et al. [8] extended the approach, showing that acoustic properties of political speech carry predictive signal for downstream behaviour. Two further results directly shape our method. Künzel [10] showed that average pitch is largely a fact about a speaker’s larynx rather than about their speech. This is the single most consequential constraint on how we measure pitch (Section 5.3). Goldman-Eisler [5] and Levelt [6] tie pausing, and more generally the moment-to-moment disruption of a planned utterance, to the mechanics of speech production. This is the literature our residual-pitch mechanism (Section 8.2) draws on directly. Laver [4] supplies the taxonomy of voice quality that defines what harmonic-to-noise ratio and steadiness of delivery are meant to capture, on both the production and the perception side.

2.3 Multimodal Political-Speech Modelling

A separate strand combines transcript features, prosodic features, and sometimes learned embeddings into supervised models that predict an outcome of interest: perceived competence, persuasiveness, or, as here, an election result. As with the debate-effects literature above (Section 2.1), this strand rarely checks whether a model has learned an outcome-associated communication pattern rather than the identity of the recurring politicians who happen to carry each label. A model can score well on held-out debates while still failing this test, if the speakers in those debates also appear elsewhere in training. We treat this as the central methodological question of the paper (Section 7) rather than a caveat appended to it.

Table 2 Studies this work builds on most directly, and the gap in each one’s design that Sections 4–7.6 address.

Study	Modality	Unit	Task	Finding	Gap addressed here
Benoit [1]	Content/text	Debate	Effects on voters	Debates shift information voters use	No delivery layer; single-cycle scope
Rosenberg & Hirschberg [7]	Acoustic	Utterance	Perceived-charisma correlates	Pitch, intensity, pausing correlate with charisma	No election outcome; no cross-speaker holdout
Gravano et al. [8]	Acoustic	Speech sample	Predict downstream behaviour	Acoustic features carry predictive signal	No multi-decade span; speaker count unreported
Tausczik & Pennebaker [9]	Text	Word	Word-counting method	Closed-class words carry social/psych. signal	Text only; no acoustic pairing
Pennebaker [3]	Text	Word	Function-word signal	Function words outweigh content words	Same as above
Künzel [10]	Acoustic	Speaker	Pitch–anatomy relationship	Mean F0 mostly reflects the larynx, not the speech	Motivates variation-not-level measurement here
Goldman-Eisler [5]	Acoustic	Utterance	Pausing and planning	Pausing indexes planning, not just rate	No election-outcome application
Levelt [6]	Theoretical	Utterance	Production model	Planned vs. online formulation differ in execution	Motivates residual-pitch mechanism (Section 8.2)
Gibson [2]	Text	Sentence	Dependency-length cost	Longer dependencies cost more to process	One of 43 syntactic measures here
Biber [14]	Text	Corpus	Register/dimension recovery	Registers need many co-occurring features	Standard our embedding axis fails (Section 5.1)

2.4 LLMs as Annotators and Judges

Zheng et al. [11] popularised the language model as a judge of open-ended output. Wei et al. [12] showed that prompting a model to reason step by step before answering can improve performance on tasks with a checkable right answer. Whether the same holds for a subjective coding scheme, where no answer key is fully reliable, is a question we test rather than assume (Section 5). Magar and Schwartz [13] showed that a model’s apparent competence on a task can be memorisation of the evaluation material rather than reasoning over it. This concern is not hypothetical for this corpus: presidential debate transcripts are among the most reproduced texts in any model’s training data. Section 7.6 therefore measures this directly, with a contamination probe and a closed-vocabulary grounding check, rather than trusting a model’s stated reasons.

2.5 Positioning of the Present Study

Prior research suggests that linguistic and prosodic properties carry political and interpersonal information. But three things remain unclear: which differences persist across election cycles, whether they generalise beyond familiar speakers, and whether LLM judgments reflect evidence-based reasoning rather than recognition of historically prominent transcripts. This paper answers all three for one corpus, under one shared statistical and measurement-validity protocol (Section 4). Table 2 summarises the ten studies this design draws on most directly and what each one’s design could not do.

3 Corpus Construction and Data Quality

3.1 Corpus Scope and Inclusion Criteria

The corpus covers every televised general-election presidential debate held in the United States between 1988 and 2020: 24 debates across nine election cycles. This is a complete enumeration for the period rather than a sample. Vice-presidential debates are excluded because consistent transcripts do not exist for all cycles.

For each debate we collected the broadcast audio and the official transcript, and paired them by election year and debate number. Election-outcome metadata supplies one record per candidate per debate. Each record carries the candidate’s party affiliation and whether that candidate’s ticket won the election that followed. [Appendix A](#) lists every debate, its date, its candidates, and the outcome attached to each.

Two further debates, from 2024, were collected but are held out of every analysis reported here. Their speaker attribution could not be verified. Their official transcripts are paraphrased news summaries that share only 5.2% and 5.5% of their words with the audio. [Appendix I](#) documents the evidence and reports what changes if they are included.

3.2 Transcript Preprocessing

Matching text to audio and analysing language need opposite things. Matching needs a regular token stream, so we build a stripped version of the text with lower case, no punctuation, and contractions left alone. Analysis needs the opposite, because sentence boundaries, part-of-speech tags and dialogue acts all depend on capitals and punctuation. We keep both and never analyse the stripped one. Stage directions and applause markers resolve to no speaker and are dropped before sentence segmentation.

3.3 Speaker-Label Normalisation

Speaker labels vary across three decades and even within one transcript (MR. BUSH, BUSH:, THE PRESIDENT). Each is matched against the roster for that debate and resolved to one of four roles: candidate (with party and outcome), moderator, audience, or unattributed.

3.4 Audio–Text Alignment

Broadcasts include studio commentary before and after the debate itself. This commentary is not part of the debate, and it would corrupt both the audio measurements and the speaker list. We locate the start and end by finding the opening and closing formulae in the audio, and time everything from there. [Figure 1](#) summarises the full pipeline from raw audio and transcript to the aligned corpus used in every analysis below.

Every acoustic measurement needs to know which piece of audio corresponds to which sentence. This is the hardest practical step, because the transcripts are not equally faithful. Some were produced verbatim, and others were tidied for readability. Rather than assume, we measure. For each debate we run speech recognition on the

Table 3 How the transcript was matched to the audio, decided by the measured match rate before any analysis.

Treatment	Debates	Sentences	Match rate
High fidelity	22	25,693	88.4–93.3%
Reduced fidelity	2	3,979	69.7–82.3%
Total	24	29,672	—

audio and count how many words of the official transcript also appear, in order, in the recognised text. This match rate is computed before any decision is taken. It determines how the debate is handled (Table 3). Where the match is high, the official transcript is treated as authoritative. Shared words take their timings directly, and the gaps between them are filled by interpolation. Where it is lower, the same method applies with more interpolated timing.

Forced alignment was also run and worked on 20 of the 24 debates. We still used the recognition-based method throughout, for a design reason. If 20 debates took timings from one method and four from another, the corpus would carry a difference in timing accuracy that lines up with the age of the recording, and therefore with the period. That is exactly the confound the analysis is built to remove. One method everywhere avoids it. The forced alignment output is kept as an independent check (Appendix B).

Each sentence then receives measurements at four levels: linguistic, semantic, discourse and acoustic. These are detailed, tools and all, in Section 4.2.

3.5 Quality Control and Exclusions

Of the 24,193 candidate sentences, 23,109 (95.5%; 10,868 winning, 12,241 non-winning) carry complete acoustic measurements. The 1,084 missing are 846 sentences shorter than the 300 ms extraction floor plus 238 extraction errors. Missing data is even across the two outcome groups (4.3% against 4.7%), but it is very uneven across alignment treatments (2.8% against 16.5%). This unevenness is one of several reasons every statistical model below fits one term per debate (Section 4.4). Fitting one term per debate removes any difference between debates, including this one, from the comparison of interest. No two candidate sentences overlap in time, so interruptions and overlapping speech are not represented anywhere in the corpus. Appendix I lists the remaining known data-quality issues (near-duplicate sentences, zero-duration rows, extraction failures) in full.

3.6 Final Corpus Statistics and Speaker Structure

The corpus holds 29,672 time-aligned sentences (Table 4), of which candidate sentences—the analytical population—are 81.5%. Table 5 gives the breakdown by election. The two groups differ in size for two reasons: candidates get different amounts of floor time, and in 1992 three people shared the stage, two of whom lost. This does not affect the analysis, since every debate contributes both groups and all comparisons are made within debates.

Table 4 Who speaks in the corpus.

Speaker role	Sentences	Share
Candidate, winning	11,355	38.3%
Candidate, non-winning	12,838	43.3%
Moderator	5,192	17.5%
Audience	287	1.0%
Total	29,672	100%

Table 5 Candidate sentences by election cycle.

Cycle	Debates	Winning	Non-winning
1988	2	821	770
1992	3	832	2,058
1996	2	816	1,270
2000	3	1,625	1,192
2004	3	1,458	1,569
2008	3	1,189	1,302
2012	3	1,227	1,733
2016	3	2,035	1,250
2020	2	1,352	1,694
Total	24	11,355	12,838

One statistic does not appear in either table. It governs the interpretation of every result in this paper: these 24,193 sentences come from 14 **people**. Twelve of them contested a single election, so they appear in one outcome group only. This is not a minor limitation stated in passing. Sections 7 and 8.3 show in detail why it governs whether any predictive result can be trusted. Table 6 gives the structure.

3.7 Descriptive Profile of Candidate Speech

Before applying the within-debate framework of Section 4, we report a purely descriptive, pooled profile of candidate speech in the Analytical Corpus: group means and percentages with no debate or speaker controls. We include it because several of these pooled contrasts shrink or reverse once the same comparisons are made within debates and checked against speaker identity (Sections 5.1–5.3). The gap between the pooled picture and the validated one is itself informative: it shows how easily an uncontrolled comparison can mislead in this corpus. This is why we report both profiles rather than only the validated one.

Table 7 profiles sentence form, syntactic complexity and lexical composition. Declarative sentences dominate both groups (about 93% of utterances). Winning candidates use somewhat more imperatives (3.30% against 2.62%) and slightly longer sentences (14.03 against 13.60 words). They also show more clausal material, marginally deeper parses, slightly more nouns and adjectives, and slightly fewer third-person pronouns. Lexical diversity and readability are essentially identical between

Table 6 Who is in the corpus. Twelve of fourteen candidates appear in only one outcome group; two appear in both.

Speaker	Debates		Sentences	
	Win	Lose	Win	Lose
Donald Trump	3	2	2,035	1,694
George W. Bush	6	0	3,083	—
Barack Obama	6	0	2,416	—
George H. W. Bush	2	3	821	1,034
Bill Clinton	5	0	1,648	—
Mitt Romney	0	3	—	1,733
John Kerry	0	3	—	1,569
Joe Biden	2	0	1,352	—
John McCain	0	3	—	1,302
Bob Dole	0	2	—	1,270
Hillary Clinton	0	3	—	1,250
Al Gore	0	3	—	1,192
Ross Perot	0	3	—	1,024
Michael Dukakis	0	2	—	770
Total	24	24	11,355	12,838

Table 7 Descriptive profile of candidate speech in the Analytical Corpus: pooled means or percentages per sentence, with no debate or speaker controls.

	Winning	Non-winning
Declarative sentences (%)	92.86	93.57
Imperative sentences (%)	3.30	2.62
Words per sentence	14.03	13.60
Clauses per sentence	1.84	1.74
Subordinate/clausal material	1.41	1.26
Parse-tree depth	5.25	5.13
Type-token ratio	0.920	0.919
Nouns (% of tokens)	13.3	12.8
Adjectives (% of tokens)	4.9	4.5
Third-person pronouns (%)	3.9	4.2
Flesch-Kincaid grade level	6.61	6.41
Mean compound sentiment	0.086	0.070

the two groups, and mean sentiment is mildly positive for both. Among the 35 Empath topic categories, winning candidates score higher on governance- and policy-related themes (*business, economics, money, government, law, help*), and non-winning candidates score higher on *leader* and *violence*. The absolute differences are small throughout.

In the discourse layer, statements dominate as expected (93.2% of candidate sentences carry the statement tag; informational statements 9.6%, explanations 6.2%, personal stance 5.5%, patriotic appeals 4.8%). Table 8 lists the categories with the largest pooled win/non-win frequency differences. Winning candidates use more interactive and explanatory acts (imperatives, explanations, agreement, rejection, direct

Table 8 Discourse-act and rhetorical categories with the largest absolute pooled frequency differences between outcome groups (percentage of candidate sentences; uncontrolled for debate or speaker).

Category	Type	Winning	Non-win.	Δ
Patriotic appeal	Rhetorical	4.29	5.18	-0.89
Imperative/directive	Dialogue act	3.30	2.62	+0.68
Explanation	Dialogue act	6.44	5.91	+0.53
Agreement	Dialogue act	1.16	0.77	+0.40
Historical reference	Rhetorical	0.84	1.21	-0.36
Emotional appeal	Rhetorical	1.51	1.81	-0.30
Rejection	Dialogue act	0.85	0.62	+0.23
Yes-no question	Dialogue act	0.55	0.74	-0.20
Direct answer	Dialogue act	1.11	0.93	+0.18
Unity framing	Rhetorical	1.50	1.64	-0.14

answers). Non-winning candidates use more rhetorical framing (patriotic appeals, historical references, emotional appeals, unity framing). Section 5.2 shows that several of these categories are dominated by single lexical triggers, so we read this table as descriptive of the annotation layer rather than as a validated behavioural difference.

Acoustically, the same pooled comparison shows winning candidates pausing more (mean total pause time 0.402 s against 0.290 s per sentence; 1.02 against 0.77 pauses) at a slightly higher syllable rate (5.64 against 5.51 syllables/s). Section 5.3 shows that this pausing difference does not survive removing a single speaker: exactly the kind of gap between the pooled and validated pictures noted above.

4 Analytical and Experimental Framework

4.1 Outcome and Unit of Analysis

Every candidate-debate record carries a binary outcome label, defined as follows:

$$w_{d,s} = \begin{cases} 1, & \text{ticket } s \text{ won the election after debate } d, \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

We use three units of analysis. *Sentence level* is the finest grain, used for the statistical screen (Section 4.4) and for training classifiers (Section 7.2). Measurements exist at this level, but the outcome label does not. The label is assigned once per candidate-debate, so all sentences from one candidate in one debate share one label. *Candidate-by-debate* is the aggregate used wherever speaker identity itself is the variable of interest (Table 6; Section 7). *Debate-level paired prediction* asks, for a whole debate, whether the model’s aggregate score ranks the winning candidate above the opponent. Every generalisation claim in Sections 7 and 7.6 is judged at this level, because it is the only one of the three not affected by class imbalance or by dependence between sentences from the same speaker.

4.2 Broad Multimodal Measure Set

We measure each sentence at four levels: linguistic, semantic, discourse and acoustic. Together these come to 43 measures; the complete dictionary is in Appendix C. The next four subsections describe how each level is built, in the order a sentence passes through the pipeline: what it says, what it means, what it does as an utterance, and how it sounds.

4.2.1 Linguistic Representation

Sentence-level parsing and part-of-speech tagging use spaCy (`en_core_web_sm`) [15]. We classify sentence form (declarative, interrogative, imperative, exclamatory) from punctuation and dependency rules. We measure structural complexity with clause count, parse-tree depth, and mean dependency length. Clause-related counts come from the dependency relations `ccomp`, `xcomp`, `advcl`, `acl` and `relcl`. This set includes some non-finite complements, so we read the resulting count as *clausal material* rather than as a literal subordinate-clause count; Section 5.1 shows why that distinction matters once the numbers come in. Lexical measures are sentence length, mean word length, type-token ratio, the balance of function to content words, part-of-speech proportions, and separate first-, second- and third-person pronoun rates. Readability is four scores: Flesch Reading Ease, Flesch-Kincaid Grade Level, SMOG, and Gunning Fog.

4.2.2 Semantic Representation

We estimate sentence-level sentiment with VADER: compound polarity, plus separate positive, negative and neutral components [16]. We measure topic content with 35 Empath categories [17], chosen before we ran any outcome analysis, to represent politically relevant themes. We treat these as a semantic diagnostic family, not as 35 additional primary hypotheses. We represent contextual meaning with 384-dimensional Sentence-BERT embeddings (`all-MiniLM-L6-v2`) [18], used only for exploratory analysis, since no single dimension of an embedding has a direct linguistic interpretation.

4.2.3 Discourse Annotation: DAMSL and BEADS

Dialogue acts follow the DAMSL framework [19]. We complement it with BEADS, a framework of rhetorical and bias-related categories we built for political debate speech: personal stance, patriotic appeal, fear appeal, historical reference, unity framing, and others [20]. The combined scheme has 28 dialogue-act categories and 21 BEADS categories, and it is multi-label: one sentence can carry more than one tag. The full tag inventory, with definitions and worked examples, is documented separately [20]. We generate primary annotations with a transparent, rule-based tagger that combines sentence form, lexical and phrase patterns, question structure (auxiliary inversion for yes-no questions, interrogative words for wh-questions), and first-person stance markers. We also generate an independent, model-based annotation as a second measurement instrument. Section 5.2 treats agreement between the two as a

measurement-validity question, and treats neither one as human-adjudicated ground truth.

4.2.4 Acoustic Representation

Acoustic measures are average pitch, its variation, range and slope, computed both in hertz and on a semitone scale. A hertz-based measure of variation grows with average pitch automatically, so the semitone scale is needed for a fair comparison across speakers. Acoustic measures also include loudness and its variation; three measures of the voice source, including harmonic-to-noise ratio computed by two independent tools; silent-pause count, total duration and average duration; and speech and articulation rate in words and syllables. Appendix C lists every tool, version and parameter setting behind these four subsections.

4.3 Targeted Pitch-Contour Measure Set

Section 5.3 shows that narrower semitone pitch range is the one acoustic property that survives every validity check in the broad screen. Every measure up to that point treats pitch variability as a single number per sentence—a standard deviation, or the width of a percentile band. Such a number answers *how much* a speaker’s pitch moves and says nothing about the *shape* of that movement. We therefore extract the pitch contour itself and build ten further measures of its shape and dynamics, as a targeted follow-up on the one construct already established by the broad screen.

For each candidate sentence we recover the frame-level semitone-scaled pitch contour. This is the same quantity summarised by the surviving broad-screen measure, but recovered before it is collapsed to one number. We extract it using the openS-MILE eGeMAPSv02 low-level descriptor `F0semitoneFrom27.5Hz_sma3nz` at its native 10 ms frame rate, extracted independently per sentence clip to match the original per-clip procedure exactly. We validate this extraction against the already published semitone functionals for 1,000+ sentences and find the two agree to within floating-point rounding (mean absolute difference 0.0006). A sentence needs at least 10 voiced frames (100 ms of voiced speech) to be included. This is a floor rather than a guarantee of reliability, so we additionally assess reliability by stratifying on duration after the fact, the same way we do for voice clarity in Section 5.3.

From the voiced frames of each sentence we compute ten measures in two groups. *Distributional shape* measures are order-independent. *Skewness* indicates whether the distribution leans toward higher or lower pitch, with a longer tail on one side. *Kurtosis* indicates whether frames cluster near a central value with occasional large excursions, or instead spread more evenly across the range. The *interquartile-to-range ratio* is the width of the middle half of the distribution relative to its 20–80 percentile span. *Dynamics* measures are order-dependent: they are computed only across pairs of consecutive voiced frames, so no estimate spans a silent or unvoiced gap. *Pitch velocity* is the mean and standard deviation of the frame-to-frame semitone change. *Pitch acceleration* is the mean absolute second difference. *Turning points per second* is how often the contour changes direction. *Total variation* is the summed absolute frame-to-frame

movement normalised by voiced duration. Finally, we fit a linear trend to each sentence’s contour and remove it; from this we compute the *declination slope* of that trend and the *declination-residual standard deviation*, which is the spread of whatever pitch movement is left once the sentence’s overall rise or fall is subtracted out. We evaluate all ten with the statistical design of Section 4.4, applying Benjamini–Hochberg correction at $q = 0.05$ across this family of ten measures. This family is kept separate from the broad-screen family of 43.

4.4 Within-Debate Statistical Comparison

Every analysis in this paper, broad screen and targeted follow-up alike, uses one statistical design, described once here.

Comparing within debates. For each measure y and sentence i , with $w_i = 1$ for winning candidates, we fit

$$y_i = \beta w_i + \sum_d \alpha_d D_{i,d} + \varepsilon_i, \quad (2)$$

where $D_{i,d}$ marks sentence i as belonging to debate d . One term per debate removes everything that varies between debates, so β is the difference *inside* debates. It is free of differences in period, format, equipment, moderator, and topic. We report $\beta/\sigma_{\text{within}}$, the difference in standard deviations.

Uncertainty and p -values. We compute cluster-robust standard errors, treating each debate as a cluster and, separately, treating each speaker as one. Every effect carries a 95% confidence interval. Our primary p -value is an exact randomization test: we flip the winner label within each debate and recompute the effect. With 24 debates there are $2^{24} = 16,777,216$ possible flips, and we enumerate all of them.

Correcting for multiple measures. With 43 measures, about two would clear $p < 0.05$ by chance alone. We control the false discovery rate at $q = 0.05$ with the Benjamini–Hochberg procedure [21]: we sort the p -values smallest first and compare the k -th against $k \times 0.05/43$.

Asking whether a result would repeat. We pool the 24 debates with a random-effects model. We report how much the effect varies between debates (I^2), a 95% confidence interval for the average, and a 95% *prediction interval* for a new debate. I^2 is the percentage of the total variation across debates that reflects real differences between debates rather than sampling noise; 0% means the debates agree up to noise, and values near 100%, which is what we find throughout this paper, mean the effect size itself swings from debate to debate. The confidence interval answers what the average difference was across the debates we have. The prediction interval answers whether we would see the difference at all in a debate we have not.

4.5 Measurement-Validity Protocol

Significance and multiple-comparison correction are not sufficient grounds to call a measure reliable. We therefore require every candidate measure in this paper to clear seven further checks before it is reported as reliable rather than merely detected:

1. **Speaker coupling.** Does the measure correlate with a speaker’s own habitual pitch, across speakers and within a single speaker’s own debates (Sections 5.3, 6)?
2. **Cross-implementation agreement.** Does it agree across independent tools measuring the same quantity (Praat versus openSMILE; Section 5.3)?
3. **Duration reliability.** Is it reliable across the range of sentence durations actually present in the corpus (Table C3)?
4. **Alignment-treatment stability.** Does it hold sign under both the high- and reduced-fidelity alignment treatments (Table 3)?
5. **Leave-one-speaker-out stability.** Does it retain direction when each speaker in turn is removed and the effect recomputed?
6. **Cross-debate recurrence.** Does it appear in a majority of the 24 debates individually, not only in the pooled average?
7. **Prediction-interval support.** Does its 95% prediction interval for a new debate exclude zero?

Section 5.3 reports the largest observed effect in the paper failing checks 1–2. Section 6 reports the primary finding passing checks 1–6 and failing check 7. We treat this as the paper’s central qualification rather than a caveat appended to it.

4.6 Generalisation Protocol

We use two evaluations throughout, and they answer different questions. **Debate held out:** the debate under test is absent from training, but its speakers’ other debates may remain. This is the standard setup in most published work. **Speakers held out:** every debate containing either of the test debate’s speakers is absent from training, so neither candidate’s voice appears anywhere in the model. We treat the second as the primary evaluation of generalisation in this paper. We also report the first alongside it, because the gap between the two is itself the result we report in Section 7.

4.7 LLM Evaluation Protocol

For each of the 24 debates we build an anonymised record. Every candidate sentence has its names, party, and the date removed. Each candidate is relabelled *Candidate A*, *B*, or *C*, in an order that is re-randomised on every repeat. We ask the model (Anthropic’s `claude-opus-4-8`; full specification in Appendix G) which candidate went on to win that year’s election. We ask this under two conditions, both built from the same 34 of the 43 measures used in Section 7.1 (those with complete values on enough rows). In the **transcript** condition the model sees only the anonymised candidate turns, in order, and nothing else. In the **descriptors** condition it sees the 34 within-debate standardised measures, averaged per candidate, as a table, with no transcript text at all. We also run a third condition with both together. It agrees with the transcript-only condition on all 24 debates, so we report it only in Appendix F.

Every call also requires a short explanation and up to five cited reasons. Each reason is drawn from a closed vocabulary: the 34 measure names for the descriptor condition, or eight qualitative themes for the transcript condition. This closed vocabulary means a citation can be checked by machine against the real data, rather than read and judged

Table 9 Sentence-structure measures:
within-debate effect size, in standard deviations.
Positive means higher among winning candidates.
None survives Benjamini–Hochberg correction.

Measure	Effect
Subordinate clauses (clausal material)	+0.093
Clause count	+0.086
Sentence length	+0.034
Parse depth	+0.029

by hand. We run two further checks alongside the prediction itself. A **contamination probe** sends the model the same anonymised input, with no prediction question at all, and asks it to identify the specific debate. A **grounding check** compares every cited measure against the real standardised value for that debate. We score each debate three times per condition, re-randomising the candidate letters each time.

5 Broad Multimodal Screening

This section answers RQ1: which properties differ, within debates, between winning and non-winning candidates, across 43 measures at four levels. It is also where the pooled, uncontrolled comparison of Section 3.7 meets the validated one. Several of the contrasts in Tables 7 and 8 shrink, disappear or reverse once the same measures are compared within debates. We flag each case as it comes up below. Figure 2, at the end of this section, plots all 43 measures side by side; the prose below walks through what is in it, level by level, and Table 11 gives a one-line verdict on each of the three acoustic patterns that stand out in it.

5.1 Linguistic and Semantic Results

Before reading any result from the subordination counter, one property of it must be stated. It accepts dependency relations that admit non-finite complements, and it reports 1.41 subordinate clauses out of 1.84 total in a fourteen-word sentence. This is too high to be real subordination, so we read this measure as “clausal material” rather than subordination.

Sentence structure. The pattern here is weak and points one way. Table 9 gives the four measures side by side: winning candidates produce slightly more clausal material on every one of them, but none of these effects survives correction, and every confidence interval includes zero. The effects are also two to eight times larger in the reduced-fidelity treatment, which places most of them in two debates from one cycle rather than across the corpus.

Words and semantics. Adjective rate is the strongest single text measure, at +0.062 ($p = 0.0036$). It misses correction by a hair. It clears correction in any family of 41 measures or fewer, but not in 43. We report it without explanation, since we had no prior expectation about adjectives. This is the one measure in this section where the pooled and the validated comparison agree in direction. Table 7 already showed

winning candidates using more adjectives (4.9% against 4.5% of tokens), with no debate or speaker controls at all. The within-debate effect points the same way, even though it does not survive correction. Pronouns do not separate the groups at all: second person -0.048 , third person $+0.009$, first person -0.003 . Third person is the case where pooled and validated comparison disagree. The raw pooled numbers of Table 7 put winning candidates *lower* (3.9% against 4.2%). This is the opposite sign from the within-debate effect above, which is itself indistinguishable from zero. Neither number supports a real difference, but the sign flip between them is a useful warning: a pooled comparison this small can point either way depending on which debates happen to carry more of each group. We record the pronoun null on its own, because in a debate *you* nearly always means the opponent: if these two groups differ, it is not in how much they address each other. Readability and sentiment do not separate the groups either ($+0.030$, $p = 0.27$).

The embedding axis. We draw the line between the two groups’ average sentence vectors and sort the corpus along it. This produces a striking result: the 50 highest-scoring sentences split 40 to 10 between winning and non-winning candidates, a stronger separation than anything else in this section. A closer look at these sentences shows that this split does not, in fact, reflect a separation between the groups. Among the 200 highest-scoring sentences, 47.5% contain *she*, *her* or *hers*, against 1.9% in the corpus overall. Of the top 50 sentences, 32 come from one candidate and 29 from the 2016 cycle. The axis separates talking about a female opponent from talking about past presidents, in one election. It is not a sentence-length axis either: the correlation is $+0.09$, and removing length leaves the poles unchanged. We report this as a diagnostic of what a single difference-of-centroids direction can produce by accident, not as a finding: following Biber [14], a register dimension would have to be recovered from many co-occurring features across texts.

5.2 Discourse-Measure Validity

This layer produced no usable result. The reason lies in the instrument, not in the underlying construct. Table 8 already showed pooled, uncontrolled frequency differences of up to 0.89 percentage points on exactly these categories, with patriotic appeal, historical reference and unity framing all higher among non-winning candidates. The following audit is why we do not read that table as a behavioural finding. We check what fraction of each category’s hits come from a single string. Several of the rule-based detectors turn out to be keyword matchers rather than concept detectors. Unity framing and patriotic appeal are dominated by the strings *United States* and *America/American*. Historical reference is dominated by a bare four-digit year. Fear appeal is dominated by unbounded substring matches. Table 10 audits the four fixable categories directly. Fixing the rules removes 94–97% of what these categories had counted. What survives appears in under 0.5% of sentences, with differences no larger than 0.014.

A second, independent labelling by a large language model agrees with the rule-based one at chance on the interpretive, rhetorical categories (Krippendorff’s $\alpha = +0.012$), and only modestly better on the more grammatically observable dialogue acts ($\alpha = +0.199$). A chain-of-thought variant of the same prompt does not close this

Table 10 Audit of four rule-based discourse detectors. “Sole trigger” is the share of hits produced by the listed string alone. “Repaired” is the count after anchoring the rule to that string at a word boundary, with the effect recomputed on what survives.

Category	Orig. hits	Dominant trigger	Sole trigger	Repaired (hits, effect)
Unity framing	396	“United States”	80.1%	69, -0.004
Patriotic appeal	1,235	“America”	88.0%	115, -0.014
Historical ref.	265	four-digit year	63.4%	165, -0.028
Fear appeal	540	substring match	30.9%	373, $+0.007$

gap. We draw no conclusion about rhetorical strategy from this layer anywhere in this paper. The full trigger audit, category-level agreement table, and chain-of-thought comparison are in Appendix E (Table E4).

5.3 Acoustic Results

Following Künzel [10], we treat average pitch as weak evidence when comparing different people. We treat variation as more meaningful, since variation describes what a speaker does rather than what their larynx is. Any measure that correlates with a speaker’s own average pitch is therefore suspect, and we test each one for this correlation.

The two measures that survive. The semitone pitch range between the 20th and 80th percentiles averages 4.51 for winning candidates and 4.91 for non-winning ones: winners move their pitch across a narrower band. The difference is -0.226 standard deviations, confidence interval $(-0.332, -0.120)$, randomization $p = 0.00017$, present in 21 of 24 debates. Dropping any one speaker leaves it between -0.308 and -0.169 , always the same sign. It holds in both alignment treatments (-0.229 and -0.206). It barely correlates with a speaker’s own average pitch ($r = +0.11$). Correcting for average pitch makes it *stronger* (-0.267). The second measure, pitch variation relative to average pitch, is larger (-0.332 , $p = 8.8 \times 10^{-6}$, 20/24) but less clean. It correlates with a speaker’s average pitch at $r = -0.49$, and it collapses to $+0.021$ in the reduced-fidelity treatment. We treat the percentile range as primary and the ratio as corroborating, a choice made on measurement quality rather than on p -values. Both measures survive Benjamini–Hochberg correction and would survive the stricter Bonferroni threshold.

Voice clarity: the largest effect, and what it measures. The biggest single effect in the study is harmonic-to-noise ratio: $+0.461$ on one tool, $+0.315$ on the other. At face value, winners have clearer voices. The validity checks say we cannot tell what it measures. Across the fourteen speakers, average pitch correlates with average harmonic-to-noise ratio at $r = +0.776$ on one tool and $+0.896$ on the other. Ranking the speakers by voice clarity nearly reproduces the ranking by pitch, most likely because a fixed 60 ms analysis window gives a low-pitched voice fewer cycles to work with, and because television transmission cuts low frequencies. The two tools then behave differently under a within-speaker check. In 51 speaker-by-debate cells, the

Table 11 The three acoustic patterns of this section, and why only one survives every check in Section 4.5.

Pattern	Effect	Debates	Verdict
Semitone 20–80 range	−0.226	21/24	Survives every check; the paper’s primary broad-screen result
Semitone SD/mean	−0.332	20/24	Corroborating only; coupled to the speaker’s own habitual pitch
Voice clarity (both tools)	+0.461 / +0.315	13/24	Fails; tracks the speaker’s larynx, not behaviour
Pausing	+0.14 to +0.20	14/24	Fails; reverses when one speaker (Obama) is dropped

correlation collapses to a median of +0.083 for one tool. For the other tool, it stays at +0.566 and is positive in all 51 cells. Two standard tools for the same quantity should not disagree this way if both are measuring voice quality. We take this as the clearest single illustration in the paper of Section 4.5’s checks 1–2: same family, same corpus, same statistics as the surviving pitch-range measure, and this one fails both.

Pausing. Winners pause more on every version of the measure (+0.14 to +0.20 across five variants), consistent across alignment treatments and duration bins. The speaker check rules this out. **All six pause measures change sign when Barack Obama is removed**, from +0.196 to −0.021 for pause proportion. His three 2012 debates alone carry effects of +0.96 to +1.25. The measure itself is sound. The finding is a description of one person, not of winning candidates in general.

Table 11 puts the three patterns side by side. Only the semitone percentile range clears every check; the other two look just as strong, or stronger, on the numbers alone.

Figure 2 shows all 43 measures together.

6 Residual Pitch Control: The Primary Finding

This section answers RQ2: what specific property of the pitch contour accounts for the strongest and most robust acoustic difference.

6.1 From Narrower Pitch Range to Residual Movement

We build the result in four steps. First, the broad screen (Section 5.3) shows winning candidates use a narrower semitone pitch band. Second, that number alone does not say what kind of movement produces the narrower band: a flatter overall melody and a steadier local wobble on top of the same melody would both shrink it. Third, the frame-level measure set of Section 4.3 asks the question directly, by removing each sentence’s own linear rise or fall and measuring what is left. Fourth, what is left is smaller for winning candidates: a specifically *local*, moment-to-moment difference in pitch control, not a difference in overall prosodic sweep. Figure 3 illustrates the distinction with two synthetic contours that share one sentence-level sweep and differ only in residual movement.

Table 12 The ten pitch-contour measures. Effect is $\beta/\sigma_{\text{within}}$. Dir. is debates in the pooled direction of 24. I^2 is in per cent. LOSO stable means the effect keeps its sign when each of the fourteen speakers is dropped in turn (leave-one-speaker-out). PI is the 95% prediction interval. Ordered by exact randomization p .

Measure	Eff.	p_{perm}	Dir.	I^2	LOSO stable	BH
Declination-residual SD	-0.242	0.0005	20/24	96	yes	•
Skewness	-0.265	0.0014	19/24	97	yes	•
IQR / 20–80 ratio	-0.053	0.0036	16/24	24	yes	•
Turning points / sec	+0.039	0.431	12/24	93	no	–
Declination slope	+0.014	0.492	14/24	65	no	–
Pitch acceleration	-0.023	0.667	14/24	92	no	–
Pitch velocity (mean)	-0.025	0.668	13/24	93	no	–
Total variation	-0.025	0.668	13/24	93	no	–
Kurtosis	-0.019	0.796	14/24	97	no	–
Pitch velocity (SD)	+0.013	0.881	14/24	97	no	–

6.2 Primary Result

Table 12 and Figure 4 show that three measures clear correction and seven do not. **Declination-residual standard deviation is the cleanest result in the paper.** Winning candidates have less pitch movement left over once each sentence’s own linear rise or fall is removed: effect -0.242 , exact randomization $p = 0.0005$, present in 20 of 24 debates (sign-test $p = 0.0015$), sign-stable across all fourteen leave-one-speaker-out refits. It passes the two checks that separated the broad screen’s two surviving measures from each other (Section 5.3). Correlation with a speaker’s own mean pitch is $r = -0.06$ across speakers and $r = +0.23$ within speaker-by-debate cells. It is uncoupled by either reckoning, unlike the confounded voice-clarity measure. Rather than shrinking with sentence length, the way a noise artefact would, it *grows*: from -0.17 in sentences under a second to -0.53 in sentences over six. It holds sign in both alignment treatments, more weakly in the reduced-fidelity group (-0.26 against -0.08) but without reversing.

6.3 Secondary Pitch-Shape Findings

Skewness survives the same correction, but it fails the same check that demoted the broad screen’s secondary pitch measure. It correlates with a speaker’s own mean pitch at $r = -0.29$ across speakers and $r = -0.24$ within cells. It also reverses sign under alignment treatment (-0.31 against $+0.08$). By the same standard applied throughout, we report it as corroborating rather than clean. *The interquartile-to-range ratio* has the lowest heterogeneity of any measure in this paper ($I^2 = 24\%$ against 94–97% elsewhere). It is stable across duration bins and alignment treatment. Its per-debate sign count, however, is the weakest of the three (16/24, sign-test $p = 0.15$). It also shows the same across-speaker pitch coupling as skewness ($r = -0.26$). We report it as secondary: consistent in magnitude whenever it appears, but appearing less consistently than either of the other two.

6.4 What the Null Results Rule Out

Pitch velocity, acceleration, total variation and turning-point rate—every measure of how quickly or how erratically the contour moves—show no difference between the groups, with I^2 above 90% throughout and prediction intervals spanning zero by a wide margin. Kurtosis and the declination slope itself show none either. Taken with the positive result above, this rules out an entire class of explanation: it is not that winning candidates move their pitch faster, more smoothly, or less erratically. It is specifically that the *spread* of where their pitch sits is smaller. More precisely, it is the spread that remains after removing each sentence’s own trend that is smaller. The data therefore support a distributional account of the effect rather than a kinematic one.

7 Generalization, Speaker Confounding, and Contamination

This section answers RQ3 and RQ4 together, because they turn out to be the same finding reached twice. Three classical models built on the measures of Section 5 look like they predict the winner. A large language model given the raw transcript looks like it does too. But all four are actually detecting who is speaking, or which debate this is, not how winners communicate. We treat this as a central result rather than a limitation appended to one.

7.1 Interpretable and Nonlinear Models

If text and audio carry different information, combining them should carry more than any single measure. We use two models that make different assumptions about how measures combine: a linear discriminant, which uses the full covariance structure, and a decision forest, which imposes no shape on the data at all. Both models use the same 34 of the 43 measures, keeping only those with complete values on enough rows, over 23,082 sentences. We standardised each measure within debate first, so only within-debate contrasts contribute:

$$z_{ij} = \frac{x_{ij} - \mu_{dj}}{\sigma_{dj}}, \quad (3)$$

We also fit a simple weighted sum: each measure is weighted by how well it separates the groups in training debates, then summed with no learned covariance. We retain it only as a baseline row in Table 15, since it never outperforms either structured model. Full method and results for principal component analysis are in Appendix F. This analysis asks whether the outcome happens to line up with the largest-variance directions in the same 34 measures, without ever seeing the labels. In short, it does not: the largest components are a sentence-length axis and a speaking-rate axis, and neither separates the groups. Speaker identity explains 83–87% of a candidate’s position on the two largest components.

Table 13 Top ten measures by permutation importance in the decision forest, under both holdout schemes. Nine of the ten are acoustic under either scheme.

Measure	Debate held out	Speakers held out
1 Voice clarity (openSMILE)	0.0308	0.0546
2 Semitone pitch SD/mean	0.0230	0.0526
3 Pitch SD (Hz)	0.0069	0.0092
4 Pause duration	0.0041	0.0033
5 Loudness	0.0040	0.0057
6 Semitone 20–80 pitch range	0.0036	0.0136
7 Pitch range (Hz)	0.0033	0.0056
8 Voice clarity (Praat)	0.0017	0.0106
9 Mean pitch	0.0014	—
10 1st-person pronouns	0.0009	—
Best text measure: 1st-person pronouns, 0.0009—about 3% of what the top acoustic measure contributes.		

7.2 Feature Importance

We fit a random forest of 400 trees, maximum depth 8, minimum 50 sentences per leaf, drawing $\sqrt{34}$ measures at random for each split, class weights balanced. Importance is permutation-based: after fitting, each measure is shuffled in the held-out data, and we record how far the area under the curve falls. This method is more trustworthy than split counts, which favour measures with many distinct values. We fit it twice: once holding out one debate at a time, and once holding out every debate containing either held-out speaker.

Table 13 and Figure 5 show that nine of the top ten measures are acoustic under either holdout scheme. The best text measure contributes about three per cent of what the top acoustic measure does. The forest leans hardest on the two measures already shown to be compromised: voice clarity and semitone pitch variation, both tied to the speaker’s own voice (Section 5.3). The measure that survives every validity check ranks only sixth. The ranking barely changes when speakers are held out. What changes, in the next section, is whether the ranking is worth anything.

7.3 Debate Held Out Versus Speakers Held Out

We use the same random forest and the same two holdout schemes to answer the practical question directly: given a sentence, or a whole debate, how often can a model say which candidate went on to win?

Alongside accuracy, Table 14 reports the area under the curve (AUC). AUC has a direct reading: take one sentence from a winning candidate and one from a non-winning candidate, both at random, and AUC is the probability the model scores the winning one higher. A model that has learned nothing scores 0.5; one that always gets the pair right scores 1.0. Unlike accuracy, AUC does not depend on group sizes, which matters here because the two outcome groups are unequal in size. 0.5 is the right baseline only when the held-out items are independent, a condition that fails

Table 14 How often the forest identifies the winning candidate. Always guessing the larger group scores 0.530.

	Debate held out	Speakers held out
<i>Sentence level</i> (23,082 sentences)		
Accuracy	0.596	0.518
Area under curve	0.630	0.523
<i>Debate level</i> (24 debates)		
Debates called correctly	17/24	12/24
p against a coin	0.064	1.00
Majority-class baseline	0.530	

Table 15 The paired test. For each debate, did the winning candidate’s mean score beat the opponent’s?

Model	Held out	Debates	p
Weighted sum	debate only	17/24	0.064
	both speakers	11/24	0.84
Discriminant	debate only	19/24	0.0066
	both speakers	10/24	0.54
Decision forest	debate only	17/24	0.064
	both speakers	12/24	1.00

for twelve of the fourteen speaker folds below, which is why we also report the paired debate-level test later in this section.

Table 14 shows that with the debate held out, sentence accuracy is 0.596 against a 0.530 baseline. Debate-level calls are 17/24, not significant against a coin ($p = 0.064$). With the speakers held out, sentence accuracy falls to 0.518, *below* the majority-class baseline. Debate-level calls are 12/24, an exact coin flip. Errors stay symmetric under speaker holdout (precision/recall near 0.49/0.54 either way), which indicates an uninformative model rather than a biased one.

The natural test of speaker generalisation is to hold out a speaker, predict their sentences, and score the result. That test cannot be run directly here. Twelve of fourteen speakers appear in only one outcome class, so within those folds there is only one class to discriminate, and the area under the curve is undefined. Pooling scores across folds is also not valid, since that would compare differently-calibrated models rather than measuring discrimination within any one of them. The valid test is paired and within debate: for each debate, does the winning candidate’s mean score beat the opponent’s? Both classes are present by construction and both are scored by the same model.

Table 15 and Figure 6 give the cleanest result in the paper. Holding out only the debate, the discriminant calls 19 of 24 correctly ($p = 0.0066$). The weighted sum and forest each call 17 of 24 correctly ($p = 0.064$). Holding out the speakers as well, the discriminant falls to 10/24, the forest to 12/24, and the weighted sum to 11/24. Three

Table 16 Repeatability. “Dir.” is debates in the same direction out of 24. I^2 is the percentage of variation between debates rather than sampling noise. LOSO is the range when each speaker in turn is dropped. PI is the prediction interval for a new debate.

Measure	Eff.	Dir.	I^2	LOSO	PI
Semitone 20–80 range	−0.226	21/24	94	[−.31, −.17]	(−.81, +.32)
Semitone SD/mean	−0.332	20/24	96	[−.44, −.27]	(−1.10, +.36)
Declination-resid. SD	−0.242	20/24	96	stable	(−.98, +.45)
Adjective rate	+0.062	17/24	59	stable	(−.10, +.23)
Voice clarity (A)	+0.461	13/24	99	—	(−1.61, +2.80)
Voice clarity (B)	+0.315	13/24	99	[+.11, +.47]	(−1.51, +2.28)
Pitch variation (Hz)	−0.228	17/24	98	[−.37, −.16]	(−1.17, +.67)
Pause proportion	+0.196	14/24	98	sign flips	(−.79, +1.03)
Articulation (syll/s)	+0.108	17/24	96	stable	(−.59, +.86)
Sentiment polarity	+0.030	17/24	74	stable	(−.20, +.26)

methods, three different assumptions about the shape of the data, one result: what looked like a model of how winners speak was a model of who the winners were.

7.4 Predictive Association Versus Recurrence

Every prediction interval in Table 16 includes zero, for every measure that survived every check up to this point. A difference can satisfy all of the following: it can be present across the whole corpus, statistically significant within debates, visible in 20 or 21 debates out of 24, and pass every check in Section 4.5. Even so, it can still tell us nothing about the twenty-fifth debate. One objection deserves an answer: perhaps 24 debates simply cannot produce a prediction interval that excludes zero. Adjective rate answers it: its between-debate variation is low ($I^2 = 59\%$), and its interval, (−0.10, +0.23), is within a factor of two of clearing. The obstacle is inconsistency between debates, not sample size. Between 59% and 99.5% of the variation in these measures is between debates rather than between candidates.

7.5 Exploratory Same-Candidate Analysis

Only two candidates were observed both winning and losing a general election. We fit the discriminant with each entirely removed, then project their debates into the space. For George H. W. Bush, the separation is complete: every debate he won scores above every debate he lost, and the model never saw him during fitting (6/6; gap +0.173). For Donald Trump the gap is +0.005, essentially nothing (4/6). One speaker’s winning and losing debates are also four years apart. The standardisation controls for that time period, but not for anything else that changed in the interim. Two cases cannot settle the exploratory question posed in Section 1.4. We report them because they are the only direct evidence the corpus contains. Full debate-by-debate scores are in Appendix H.

The textbook remedy for sentences nested inside speakers is a random effect per speaker. That remedy cannot be estimated here. Regressing the outcome label on speaker identity across the 51 candidate-by-debate cells gives $R^2 = 0.811$: **speaker**

Table 17 The language model’s debate-level accuracy (majority vote over 3 re-randomised repeats) beside how often a separate, question-free probe identifies the actual debate from the same input.

Input	Outcome accuracy	Exact debate identified
Descriptors only	13/24	0/24
Transcript only	24/24	24/24

identity explains 81% of whether a candidate won. Twelve of fourteen speakers appear in one outcome class only, so for them the two variables are the same variable. As a result, the variance of the outcome coefficient is inflated 5.3-fold, and only 36 residual degrees of freedom remain. A speaker random effect would not make the analysis better powered here. It would make it better calibrated. The honest output of a correctly calibrated model on this corpus is a very wide interval.

Section 4.7 gives the protocol for testing a large language model on the same question. What follows is the result, reached independently of the three classical models above.

7.6 Transcript-Only and Descriptor-Only Judgments

Given the anonymised transcript, the model calls the winner in 24 of 24 debates. Given only the descriptor table, with no text, it calls 13 of 24, a coin flip ($p = 0.340$). Read alone, the first number looks like the model successfully reasoning over communicative style from text. Section 7.7 shows that it is not.

7.7 Contamination Probe

We give the same model the same anonymised transcript with no prediction question attached, and ask it to name the debate. It names the specific real debate—year and both candidates—in 24 of 24 cases (Table 17). Given the descriptor table alone, it does so in 0 of 24. Presidential debate transcripts are some of the most quoted text in existence. Stripping names and dates removes the labels, but it does not remove lines like “I did not have sexual relations with that woman,” or a candidate saying “you’re likable enough, Hillary.” **The language model’s transcript-based accuracy cannot be interpreted as evidence of generalisable political-language reasoning, because the model identifies the exact debate from the same anonymised input in every case.** This is the same failure mode Section 7.3 found in the decision forest and the discriminant, though arrived at here by a completely different method. What looks like the modality that helps is the modality that leaks identity.

Figure 7 places the two results side by side. This is also the direct answer to a concern raised in Section 1.3 and revisited in Section 9.5: whether a language model can identify a debate and its outcome from anonymised material is a number to report, not a caveat to mention. Here, it is 100% for text and 0% for numbers.

7.8 Grounding of Explanations

Every prediction call also cites, from a closed vocabulary, which evidence it relied on. This lets us check the citations by machine against the real data instead of taking them on faith. Given descriptors and citing a specific measure, the citation points the right way 75.8% of the time (303/400 citations). With the transcript added, this rises to 84.3% (348/413). Read alone, the second number looks like better reasoning with more evidence. Read alongside the contamination result, a less flattering explanation fits at least as well. Once the transcript has told the model who wins, a numeric citation only has to sound consistent with an answer already fixed, not actually justify it. In the descriptors-only condition, where no such shortcut exists, the model’s most-cited measures are loudness, sentence length, voice clarity, grade level and speech rate. Its explanations read as plausible performance narratives, such as this one: “*Candidate B shows higher pitch variability, faster articulation rate, and more syntactically complex speech... which suggests a more dynamic and sophisticated delivery*”. This call was made in the 1988 debate, for a candidate who went on to lose. Every one of those measures was already found, in Section 7.2 or Appendix F, to be uncorrelated with winning or confounded with it. A fluent explanation is not the same thing as a correct one. We scan the transcript-only explanations for invented acoustic numbers, meaning any figure the model was never given. We find none in any of the 72 transcript-condition responses. On this specific, narrow check, the model does not fabricate.

7.9 Implications for LLM-as-Judge Research

Evaluating a language model’s judgment of famous political material needs at least two controls beyond accuracy: an identification or contamination probe, asked with no task question attached, and a check that cited reasons are actually consistent with the evidence presented rather than merely fluent. Neither control is specific to debates or to this model, and both are cheap to run. Section 8.6 returns to what they generalise to.

8 Discussion

8.1 What Differs Between Eventual Winners and Non-Winners

Across four levels and 53 properties, we find no robust broad linguistic, semantic or discourse-level separation between the two groups (Sections 5.1–5.2). We find one coherent pitch-control family. Within it, one measure, declination-residual standard deviation, is the strongest measurement-valid result in the paper (Section 6). The two original broad-screen pitch measures and this one describe the same underlying phenomenon at increasing precision: less overall range, less range relative to habitual pitch, and, most specifically, less local movement once each sentence’s own trend is removed.

The text-based comparison illustrates why we insist on that precision. The pooled, uncontrolled profile of Section 3.7 and the within-debate screen of Section 5.1 are the same comparison: one candidate group against the other. They are run at two different levels of control, and they do not always agree. Adjective rate points the

same way at both levels (higher among winning candidates, pooled and within-debate alike) without ever surviving correction. Third-person pronoun rate flips sign between the two. Pooled, winning candidates use it less. Within debates, the difference is indistinguishable from zero, in the other direction. The discourse-tag frequencies of Table 8 point the same way as the pooled text measures. Section 5.2 shows why we do not trust them either: several of the largest pooled differences come from detectors that fire on a single string. None of this changes what Section 6 finds. It is why we report a naive comparison at all, rather than only the validated one that survives correction.

8.2 Interpretation of Residual Pitch Stability

This analysis establishes a correlation between electoral outcome and one specific property of a pitch contour. It does not establish why that property might matter to a listener. We ran no perception experiment to find out. What follows is a plausible account, built from findings already cited in this paper. We offer it as interpretation rather than result.

Two production-side mechanisms and one perception-side mechanism point the same way. On the production side, planning an utterance in advance and executing it as one coherent unit is understood to produce a single, matched intonational gesture. Formulating a sentence while speaking it—as when a speaker reformulates a claim mid-sentence, hedges, or searches for the next word—is understood to introduce small disruptions to that gesture, as the plan is revised in real time [6]. A rehearsed answer and an improvised one are not expected to look the same on this measure. Debate answers are frequently at least partly rehearsed in advance. Separately, fine control over the laryngeal muscles that set pitch is understood to be sensitive to physiological effort and strain. Steadier voluntary control produces a cleaner pitch trajectory, and reduced control introduces the kind of small involuntary fluctuation this measure is built to detect [4]. On the perception side, the same voice-quality literature treats predictable, steady prosody as a standing cue to fluency and confidence, largely independent of what is being said [4]. If any of these mechanisms is operating here, a listener would not need to consciously notice the pitch contour to be influenced by it.

This interpretation connects directly to the factorial design we propose in Section 9.5. That design pairs independently manipulated text and prosody with listener ratings. A declination-residual measure computed on the resynthesised audio in that design would test directly whether steadier residual pitch movement raises a listener’s rating of vocal control, holding the words fixed. Nothing in the present corpus can adjudicate between the mechanisms above, or confirm that any one of them is operating rather than some fourth explanation. The corpus establishes only that the acoustic pattern exists and is not an artefact of speaker identity, sentence length, or transcription treatment.

8.3 Speaker Identity Versus “Winning Style”

Twenty-four thousand candidate sentences do not equal 24,193 independent outcome observations. They come from fourteen people. Any generalisation claim has to hold up over those fourteen people, not over the sentence count. Holding out a debate

while leaving its speakers’ other debates in training (Section 7.3) is not a held-out test of speaker identity. The gap between the two holdout schemes—19/24 to 10/24 for the discriminant, 17/24 to 12/24 for the forest, 17/24 to 11/24 for the weighted sum—is the size of that confound made visible. Feature stability under permutation importance (Table 13) does not imply transferable prediction. The same top-ranked measures survive speaker holdout, but the forest’s ability to use them does not.

8.4 What the Null Results Contribute

The null findings sharpen rather than weaken the positive one. Not sentiment. Not pronoun use, at any grammatical person. Not general syntactic complexity. Not pitch velocity, acceleration, or how often the pitch contour changes direction. Not most rhetorical categories, once we check the detectors that claimed to measure them. What remains after this many specific nulls is not “pitch matters, vaguely.” It is one measurable property of one signal. It is characterised precisely enough that a null result on a neighbouring measure (turning-point rate; kurtosis) rules out specific alternative explanations rather than merely failing to reach significance.

8.5 Implications for Multimodal Modelling

This lesson extends past presidential debates. A multimodal system trained on a small, repeatedly-observed population of subjects can improve its apparent performance by exploiting identity-bearing information, acoustic or textual, rather than the task-relevant behaviour it was meant to capture. Debate-level holdout alone will not reveal this if the same individuals recur across the held-out and training partitions. Three structurally different models (Section 7) reach this failure mode independently. A fourth, a large language model (Section 7.6), reaches the text-modality version of the same failure by a completely different route.

8.6 Implications for LLM Research

Memorisation of famous transcripts is not a remote risk for language-model evaluation on any well-known text. It is directly measurable with a question-free identification probe. Here it was total for the raw transcript and absent for numeric descriptors of the same material. Partial anonymisation (names, party and dates removed, content untouched) was sufficient to demonstrate the contamination without destroying the communicative content the transcript condition existed to test. It was not sufficient to prevent the contamination itself. That is the more important number to report. A grounding requirement—citations checkable against real data, from a closed vocabulary—separates answer accuracy from valid reasoning in a way free-text explanation cannot. We treat it as a minimum bar for LLM-as-judge use on any material a model may have seen before, not only on debate transcripts.

8.7 Relationship to Previous Research

Rosenberg and Hirschberg’s [7] expectation that pitch and pausing would matter most is broadly supported. The only broad-screen measures that survive are pitch measures.

The pausing result does not survive the speaker check, but the effect is large enough that it would have looked like a genuine finding without that check. Künzel’s [10] warning about average pitch is directly borne out by the paper’s largest and least trustworthy effect (voice clarity, Section 5.3). What this design adds, that a single-cycle or single-debate study could not, is the speaker-aware evaluation that turns “pitch range differs” into “pitch range differs, and specifically the residual component of it, in a way that does not yet predict an unseen debate.” This is a more qualified conclusion than the charisma literature’s framing, and we take it to be a more useful one.

9 Limitations and Future Work

9.1 Corpus Limitations

- **Fourteen speakers is few.** Nine elections and fourteen people can reveal only differences that are nearly universal. A null result means no evidence, not evidence of nothing.
- **Third-party candidates are not like major-party losers.** Ross Perot contributes 1,024 sentences. Dropping him and recomputing is the only evidence we have on how much he matters.
- **One language, one broadcasting tradition.**
- **Two 2024 debates are excluded** for unverifiable speaker attribution (Appendix I). Including them changes several secondary results and none of the primary ones.

9.2 Measurement Limitations

- **Some timings are interpolated.** Missing acoustic data is uneven across alignment treatments (2.8% against 16.5%).
- **Interruptions are not represented.** No two candidate sentences overlap in time, including in the 2020 debate, which was defined by people talking over each other.
- **Sentence boundaries vary across the corpus.** Every per-sentence count depends on them.
- **A quarter of sentences are too short** for reliable voice measurement (Table C3).

9.3 Inferential Limitations

- **The election result is not a measure of good speaking.** It is a shared external target. No claim here depends on winning and speaking well being the same thing.
- **This is correlation.** Winners differ from losers in incumbency, money, party position and prior office. Incumbency can be added within debates at no cost and should be the first control a follow-up adds.
- **Every prediction interval reported includes zero.** An association can be real, significant, and repeatedly observed across this corpus without yet supporting a prediction about a debate we have not seen (Section 7.4).

9.4 LLM Limitations

- **One model, one API version, one evaluation date.** Every language-model result in this paper uses `claude-opus-4-8` on 2026-08-06 (Appendix G). We tested one prompting variation (chain-of-thought against direct). It made no difference, but we cannot rule out that a different model or prompt would.
- **The training-data concern is measured, not assumed.** The contamination probe (Section 7.7) shows the model identifies the exact debate, year and both candidates from an anonymised transcript in 24 of 24 cases. This corpus cannot be absent from the training data. Its memorisation is directly measurable. Any result touching the transcript text has to be read through that fact.
- **Discourse-tag agreement here is instrument-versus-instrument, not instrument-versus-ground-truth.** We compare rule-based and model-based labels against each other (Section 5.2). Neither is validated against adjudicated human judgment. The chance-level agreement on rhetorical categories is informative about the instruments’ disagreement with each other, not proof that either is wrong.

9.5 Future Work

Two studies would extend this design most directly: an expanded observational corpus, and a controlled experiment that manipulates text and prosody independently.

An expanded observational corpus. The limit on the substantive question is the number of elections, not the amount of speech. Adding vice-presidential, primary, senatorial and gubernatorial debates would keep the format while multiplying the speakers and the contested outcomes. It would also supply many more people who both win and lose. Every check reported here carries over unchanged.

A controlled text×prosody perception experiment. In observed speech, what a candidate says and how they say it always arrive together from the same person, which is why no corpus of this kind can separate them. We propose the following design instead. Take an utterance. Rewrite the words with a language model to be more or less charismatic, holding content fixed. Resynthesise the delivery, holding words fixed. Cross the two so each utterance appears in four versions. Then have listeners rate vocal control, confidence, charisma, credibility and persuasiveness:

$$\begin{aligned}
 C = & \beta_0 + \beta_1(\text{text}) + \beta_2(\text{prosody}) \\
 & + \beta_3(\text{text} \times \text{prosody}) \\
 & + u_{\text{speaker}} + u_{\text{listener}} + \varepsilon.
 \end{aligned}
 \tag{4}$$

Now β_1 and β_2 are separately identified, because the experimenter assigned them rather than the candidate. Both random effects are estimable, because every speaker and listener appears in every condition, a property this corpus lacks. Language models can rewrite text while holding content roughly fixed. Current speech synthesis allows independent control of pitch, energy, rate and style. Together, these make this design practical now.

Two cautions carry over from this paper. The first is no longer hypothetical. Section 7.7’s contamination probe confirms directly that a language model given an anonymised transcript from this corpus identifies the debate in every case. The same check has to run on whatever text a language model rewrites for the design above, and on the rewritten output itself. It must run before any listener sees the material, not only on the original transcript. Numeric descriptors did not carry the same signal in our check. Rewritten, resynthesised speech, however, is a different object again, and it should not be assumed clean until tested. The second caution is more straightforward: these are fourteen of the most recognisable voices in America, so listeners will identify them. Whether a listener recognised the speaker should be recorded and modelled, not assumed away.

10 Conclusion

In this paper we describe a multimodal analysis of every televised U.S. general-election presidential debate from 1988 to 2020. We first compare candidate speech with a pooled, uncontrolled text-based profile. Several of its differences (adjective rate, third-person pronoun rate, patriotic and historical rhetorical framing) shrink, disappear or reverse once the same comparison is repeated within debates. For the discourse categories, they also shrink, disappear or reverse once the detectors producing them are audited. We then screen 43 linguistic, semantic, discourse and acoustic measures for a within-debate difference between eventual winning and non-winning candidates. We follow the one acoustic property that survives down to the level of the pitch contour itself. We find that eventual winning candidates show less residual pitch movement than eventual non-winning candidates, once each sentence’s own broad rise or fall is removed. The effect is -0.242 standard deviations, present in 20 of 24 debates, stable across fourteen leave-one-speaker-out refits, uncoupled from habitual pitch, and, unlike a measurement artefact, stronger rather than weaker in longer sentences. Of 53 properties measured at four levels, this is the one pattern that survives every check we ran.

We demonstrate, however, that this association does not yet support reliable prediction for an unseen speaker or a new debate: its 95% prediction interval includes zero, for a structural reason rather than a statistical one. Only fourteen people have ever taken this stage, and twelve of them did so only once. Speaker identity explains 81% of the outcome label those fourteen people carry.

We show that this same structural fact is why a decision forest, a linear discriminant, a weighted sum, and a large language model all appeared, at first pass, to predict the winner. It is also why all four stopped working once we changed the evaluation to remove speaker or transcript identity, rather than merely the debate under test. We conclude that speaker-aware validation—holding out every debate a candidate ever appears in, not just the one under test—and, for a language model, a question-free contamination probe alongside a closed-vocabulary grounding check, are not optional refinements to this kind of analysis. They are the difference between a predictive result and an artefact of who, or what text, the model had already seen.

The advantage of the speaker-aware and contamination-aware protocol used throughout this paper is that it is cheap to apply and specific to no one model: the same two holdout schemes and the same two language-model checks generalise to any multimodal study of a small, repeatedly-observed population of speakers. Limitations of the present corpus and directions for extending it are given in Section 9.

Acknowledgments

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Appendix A Debate Inventory

Table A1 Every debate in the corpus: date, candidates, and which ticket won the election that followed. Audio and official transcripts are cited by debate identifier in the deposited data statement (Appendix I).

ID	Date	Candidates	Winner
1001	1988-09-25	G. H. W. Bush vs. Dukakis	Bush
1003	1988-10-13	G. H. W. Bush vs. Dukakis	Bush
1004	1992-10-11	Clinton vs. Perot, G. H. W. Bush	Clinton
1006	1992-10-15	Clinton vs. Perot, G. H. W. Bush	Clinton
1007	1992-10-19	Clinton vs. G. H. W. Bush, Perot	Clinton
1008	1996-10-06	Clinton vs. Dole	Clinton
1010	1996-10-16	Clinton vs. Dole	Clinton
1011	2000-10-03	G. W. Bush vs. Gore	Bush
1013	2000-10-11	G. W. Bush vs. Gore	Bush
1014	2000-10-17	G. W. Bush vs. Gore	Bush
1015	2004-09-30	G. W. Bush vs. Kerry	Bush
1017	2004-10-08	G. W. Bush vs. Kerry	Bush
1018	2004-10-13	G. W. Bush vs. Kerry	Bush
1019	2008-09-26	Obama vs. McCain	Obama
1021	2008-10-07	Obama vs. McCain	Obama
1022	2008-10-15	Obama vs. McCain	Obama
1023	2012-10-03	Obama vs. Romney	Obama
1025	2012-10-16	Obama vs. Romney	Obama
1026	2012-10-22	Obama vs. Romney	Obama
1027	2016-09-26	Trump vs. H. Clinton	Trump
1029	2016-10-09	Trump vs. H. Clinton	Trump
1030	2016-10-19	Trump vs. H. Clinton	Trump
1031	2020-09-29	Biden vs. Trump	Biden
1033	2020-10-22	Biden vs. Trump	Biden

Appendix B Alignment and Preprocessing

Audio was converted to single-channel PCM at 16 kHz. For matching, both the recognised text and the official transcript were reduced to lower-case alphanumeric tokens. Punctuation was removed, contractions were left unexpanded, and numerals were kept

as written. The match rate is the share of official-transcript tokens that fall inside a matching block. Matching blocks were found with a longest-matching-block algorithm. Routing thresholds were $\geq 85\%$ and 60–85%.

Recognition used Whisper [22], which gives word-level timings. Matched tokens take those timings directly. Unmatched runs are interpolated between the nearest matched words, split by character count.

Montreal Forced Aligner [23] was run with the English ARPA acoustic model and dictionary. Default beam settings did not finish on ninety-minute recordings, so production runs used wider beams under a watchdog. It converged on 20 of 24 debates and is kept as an independent check.

Appendix C Complete Feature Definitions

Pitch, loudness and voice-quality measures come from Praat [24] through `parselmouth`. The 88-parameter set comes from openSMILE [25, 26], which also supplies the semitone pitch measures and the second voice-clarity implementation. Parsing and tagging use spaCy [15] with `en_core_web_sm`, the smallest available model. Its weakest outputs are parse depth and dependency length. Sentiment uses VADER [16], topic categories use Empath [17], and sentence vectors use Sentence-BERT [18].

A silent pause is a stretch of at least 150 ms during which loudness stays at least 25 dB below the ninetieth percentile for that sentence. The usual threshold after Goldman-Eisler [5] is 250 ms. The detector cannot see filled pauses. Whether those were transcribed at all varies across this corpus.

Table C2 names the 34 standardised measures used in Sections 7.1 and 4.7. These overlap heavily with the full 43-measure broad screen (Section 4.2). The remaining measures are named individually in the results prose of Section 5: jitter, shimmer, four readability scores, positive/negative sentiment, and the 35 topic categories, which are treated as one diagnostic family rather than as individual screened measures.

Table C2 The 34 standardised measures used in the combined and LLM analyses, by family.

Acoustic (Praat)	Acoustic (openSMILE)	Linguistic
Mean pitch	Syllable rate	Clauses
Pitch SD (Hz)	Articulation rate (syll/s)	Subordinate clauses
Pitch range (Hz)	Semitone pitch SD/mean	Parse depth
Loudness	Semitone 20–80 range	Dependency length
Voice clarity (Praat)	Voice clarity (openSMILE)	Words per sentence
Pause duration		Word length
Pause count		Function words
Mean pause length		1st/2nd/3rd-person pronouns
Speech rate		Nouns, verbs, adjectives, adverbs
Articulation rate (w/s)		Reading ease, grade level
		Sentiment, positive, negative

For the frame-level pitch-contour measures (Section 4.3), the low-level-descriptor output of openSMILE (F0semitoneFrom27.5Hz_sma3nz, 10 ms hop) was run independently per sentence clip. We first tried slicing these measures from a continuously smoothed whole-debate stream, but rejected this approach: continuous smoothing across sentence boundaries shifted the resulting statistics measurably at clip edges. Distributional measures use all voiced frames regardless of order. Dynamic measures use only differences between temporally consecutive voiced frames. Declination slope and its residual standard deviation come from an ordinary-least-squares fit of the semitone contour against frame time.

Four measurement problems are recorded for anyone reusing the corpus. *One pitch range for everyone*: the analysis range was 75–500 Hz for a corpus where average pitch runs from 102.5 to 195.9 Hz. The fixed 75 Hz floor is the likely cause of the pitch-dependence of voice clarity. *Mismatched floors*: Praat’s loudness default assumes a minimum pitch of 100 Hz against the 75 Hz used for pitch. *Voiced fraction* runs from 0.583 to 0.742 across speakers. It correlates with voice clarity at +0.674 at speaker level, and this relationship persists within a single speaker in a single debate, at a median of +0.327. *Reliability depends steeply on sentence length* (Table C3), which is why the length check in Section 5.3 had to be run first.

Table C3 Voice-clarity reliability by sentence length.
Split-half reliability is odd-versus-even sentences
within one speaker in one debate.

Sentence length	Mean clarity	SD	Split-half
0.3–1.0 s	8.79 dB	3.10	0.126
1.0–2.5 s	9.51 dB	2.42	0.476
2.5–6.0 s	9.96 dB	2.16	0.655
6.0 s+	10.30 dB	2.00	0.729
Candidate sentences: median 3.02 s, IQR (1.53, 5.71); 24.2% under 1.5 s, 13.1% under 1.0 s, 4.6% under 0.5 s.			

Appendix D Full Statistical Results

Debate effects are removed by subtracting each debate’s mean. Cluster-robust variances use a sandwich estimator with a small-sample correction. These are referred to a t distribution with $G - 1$ degrees of freedom: $G = 24$ for debate clustering and 14 for speaker clustering. With this few clusters the p -values should be read as generous rather than exact.

The randomization test enumerates all $2^{24} = 16,777,216$ ways of flipping the winner label within debates. Random-effects pooling follows DerSimonian–Laird. The prediction interval is computed as the pooled estimate $\pm t_{22} \sqrt{\tau^2 + \text{SE}^2}$. Leave-one-speaker-out recomputes the within-debate effect with one speaker’s sentences removed.

Several of the 43 measures overlap: 13 pairs correlate above $|r| = 0.7$. The effective number of independent measures is 30.0 by the Li-Ji method. Correcting at $m = 30$ instead of $m = 43$ changes nothing.

Section 7.3 defines the area under the curve (AUC) reported for the discriminant and forest, and explains why 0.5 is the right baseline only when the held-out items are independent.

The deposited version of this appendix carries the full table over all 43 measures, the treatment-split table, and all 35 topic categories.

Appendix E Discourse Annotation

The rule-based labeller assigns dialogue acts and rhetorical categories from sentence type, word and pattern matches, question syntax, and first-person stance markers. Two of its categories are copies of the sentence-type column rather than independent detectors. It returns zero for 13 codes. Table E4 shows what fraction of each category’s hits come from a single string.

Table E4 What the rule-based detectors actually fire on. “Sole trigger” is the share of hits produced by one string and nothing else. “Rep.” is the count after fixing the rule. “Eff.” is the difference after fixing.

Category	Hits	Sole trigger	Rep.	Eff.	dir.
Unity framing	396	“United States” 80.1%	69	−0.004	10/24
Patriotic appeal	1,235	“America” 88.0%	115	−0.014	13/24
Historical ref.	265	bare year 63.4%	165	−0.028	15/24
Fear appeal	540	substring 30.9%	373	+0.007	11/24
Emotional appeal	435	“children” + “families”	—	—	—
Agreement	237	“absolutely” / “exactly”	—	—	—
Rejection	178	initial “no” 93.8%	—	—	—
Statement	22,629	= declarative 100%	—	—	—
Imperative	697	= imperative form 100%	—	—	—

Fixing the four fixable rules—anchoring at word boundaries and requiring a real unity or patriotic phrase—cuts unity framing from 396 hits to 69. It also cuts patriotic appeal from 1,235 to 115. Between 94 and 97 per cent of what these categories counted was something else. What survives appears in 0.3 to 0.5% of sentences with differences no larger than 0.014.

The corpus was also labelled by a large language model through a batch API, using a prompt that assigns dialogue acts first and rhetorical categories second. The model returned a parseable answer for 100.0% of sentences. It assigned at least one dialogue act to 99.86% of candidate sentences and at least one rhetorical category to 71.05%. Table E5 gives category-level agreement. The two labellings agree best on categories readable from the grammar (questions, thanks), and worst on the interpretive ones.

Chain-of-thought variant. A natural objection to the direct prompt above is that it never gives the model room to reason. It goes straight from sentence to codes. We re-ran the labelling on a stratified sample of 3,001 candidate sentences (12.4% of

Table E5 Agreement between the rule-based and model labels over 24,193 candidate sentences. PSA (percent specific agreement) treats neither as the reference. PABAK and AC1 correct for how rare each category is.

Category	Rule	Model	PSA	κ	PABAK	AC1
Thanks	182	217	0.852	0.851	0.995	0.998
Wh-question	212	376	0.599	0.594	0.980	0.990
Polar question	146	290	0.564	0.561	0.984	0.992
Explanation	1,478	869	0.415	0.387	0.886	0.937
Personal stance	1,424	4,865	0.389	0.327	0.682	0.795
Fear appeal	540	1,674	0.237	0.210	0.860	0.923
Historical ref.	265	1,343	0.208	0.193	0.895	0.944
Patriotic appeal	1,235	282	0.190	0.174	0.898	0.946
Imperative	697	1,234	0.177	0.146	0.869	0.929
Unity framing	396	1,099	0.060	0.037	0.884	0.938
Rebuttal	3	4,653	0.001	0.001	0.616	0.767
Informative stmt.	2,475	0	0.000	0.000	0.795	0.887

Whole-set agreement (Krippendorff α , MASI):
 dialogue acts +0.199; rhetorical categories +0.012.

the corpus, proportional across all 24 debates), with one change: the output schema requires a short free-text **reasoning** field, written before the **dams1** and **beads** codes. This means the model must commit to an explanation before it commits to a label. It is the same chain-of-thought design that Wei et al. [12] showed can improve accuracy on tasks with a checkable right answer. The model, codebook, and prompt were otherwise the same (Appendix G). It does not help. Figure E1 gives Krippendorff’s α against the rule tagger for both prompting styles, recomputed on the identical sample. Dialogue acts move from +0.167 direct to +0.169 with reasoning. Rhetorical categories move from −0.030 to −0.002. Both changes are within sampling noise of each other, and both are the same chance-level result as the full-corpus comparison above. Chain-of-thought and direct prompting agree with *each other* at $\alpha = +0.564$ for dialogue acts and +0.536 for rhetorical categories. This is moderate agreement, well short of the near-unanimity two runs of a deterministic rule tagger would show. Both prompting styles remain equally uncorrelated with the rule-based labels. Reasoning also makes the model slightly more conservative: it assigns a dialogue act to 98.20% of the sample and a rhetorical category to 66.61%, against 99.83% and 70.68% for direct prompting on the identical sample. The instrument’s problem was never that the model failed to explain itself. The problem is that the categories it is asked to detect are not observable from either the grammar or an explanation. Asking for one does not manufacture the signal.

Appendix F Supplementary Multimodal Models

Weighted sum. Each measure gets a weight equal to how well it separates the groups in training debates T :

$$w_j = \frac{1}{|W_T|} \sum_{i \in W_T} z_{ij} - \frac{1}{|L_T|} \sum_{i \in L_T} z_{ij}, \quad (\text{F1})$$

The weight vector is scaled to unit length. Each sentence is then scored as $S_i = \sum_{j=1}^{34} w_j z_{ij}$. Weights are learned on 23 debates and scored on the twenty-fourth, rotating through all of them.

Table F6 Combining measures, against the two measures that survived on their own.

	Single measures		Composite (34 measures)	
	Semitone 20–80 range	Semitone SD/mean	hold out debate	hold out speakers
Mean per-debate effect	−0.243	− 0.370	+0.390	+0.070
SD across debates	0.289	0.358	0.603	0.519
Debates in direction	21/24	20/24	17/24	11/24
Sign-test p	0.0003	0.0015	0.064	0.84

On held-out debates, the weighted composite reaches +0.390 standard deviations, a 60% improvement on the semitone percentile range alone. But it is markedly less consistent (SD across debates 0.603 against 0.289). Under speaker holdout it falls to +0.070. A single measure needing no training is more reliable than a weighted combination of 34.

Discriminant loadings. Averaged to the 51 speaker-by-debate cells, the discriminant separates winning from non-winning cells at an area under the curve of 0.907. Table F7 reports the correlation between each measure and the discriminant score. Correlations are more stable than raw weights when measures are correlated.

Table F7 What the discriminant leans on. Positive means higher among winning candidates.

Measure	Loading
Voice clarity (openSMILE)	+0.628
Semitone pitch SD/mean	−0.587
Voice clarity (Praat)	+0.462
Pitch SD (Hz)	−0.422
Semitone 20–80 pitch range	−0.410
Pitch range (Hz)	−0.317
Articulation rate (syll/s)	+0.235
Mean pitch	+0.189
Pause duration	+0.186
Articulation rate (words/s)	+0.181

Every measure in the top ten is acoustic. Nothing syntactic, lexical or semantic appears until well below. The highest-loading measure is the one shown in Section 5.3 to track the speaker’s own pitch at $r = +0.78$ to $+0.90$. Consistent with that, 81.0% of the variance in the cell scores is explained by which speaker the cell belongs to.

Principal component analysis. PCA asks whether the outcome happens to line up with the largest-variance directions among the 34 measures without using the labels at all.

Table F8 Principal components of the 34 standardised measures.

	% of variance	Separation of groups	Dominant measures
PC1	20.9	+0.006	Sentence length, grade level
PC2	10.6	+0.136	Articulation rate, speech rate
PC3	7.9	+0.203	Pitch SD, reading ease
PC4	6.8	+0.070	Mean pitch, voice clarity
PC5	6.4	+0.359	Voice clarity (both tools)
PC6	5.9	+0.044	Sentiment

12 components needed for 80% of variance.
At cell level, speaker explains 83.6% of PC1 and 86.6% of PC2 positions.

PC1 (20.9% of variance) is a sentence-length dimension separating the groups at +0.006 standard deviations—nothing. PC2 is speaking rate. The component that best separates the groups is PC5, at +0.359, accounting for only 6.4% of the variance. A hand-built weighted sum, a supervised discriminant, and an unsupervised decomposition that ignores the labels entirely all reach the same place: what structure exists in these 34 measures is mostly about who is speaking.

LLM “both” condition. Given the transcript and the descriptor table together, the model calls 24 of 24 debates correctly. This agrees with the transcript-only condition (Section 7.6) on all 24 debates. Adding numeric descriptors to a transcript the model can already place in history changes nothing, because the transcript had already settled the answer.

Appendix G LLM Methods and Reproducibility

This appendix specifies every language-model call in the paper: the original discourse tagging and its chain-of-thought variant (Appendix E), and the cross-modal winner prediction with its contamination probe (Sections 4.7, 7.6). The goal is that another researcher could reissue every call from this description alone.

Model. All components use the same model, **claude-opus-4-8**, accessed through Anthropic’s Messages API. Anthropic does not publish a parameter count for this model, so none is reported here. Reporting one would imply a precision we do not have. Every call uses structured (**json_schema**) output: the model’s response is constrained at generation time to match a schema we specify, rather than parsed after the fact from free text. All calls were issued on 2026-08-06.

Sampling. This model deprecates the **temperature** parameter. The API rejects a request that sets it. Repeated calls therefore draw on whatever sampling variability the API applies by default, not on a caller-chosen temperature. Where the paper reports a repeat count, that is the only lever available for measuring stability.

Discourse tagging. System prompt, codebook and full JSON schema are reproduced verbatim in `Sahana Project/tagging_prompt.md` (direct) and `Sahana Project/cot_tagging_prompt.md` (chain-of-thought). Both send 10 sentences per request, batched for throughput, with the system prompt and codebook cached (`cache_control: ephemeral`) across requests. The only schema difference is one field, `reasoning`, which the chain-of-thought variant requires before `dams1` and `beads`. Structured output fills schema fields in the order they are declared. This forces the free-text reasoning to be generated before the codes rather than after them.

Cross-modal winner prediction. System prompt, both JSON schemas (descriptor-citing and theme-citing), the contamination-probe prompt and schema, and the closed evidence vocabularies are reproduced verbatim in `Sahana Project/crossmodal_prompt.md`. One call is issued per debate, per condition, per repeat. Each repeat uses a freshly randomised Candidate-A/B/C letter assignment, shared across conditions within that repeat so predictions are comparable trial-for-trial. The contamination probe issues one further call per debate per input type, with no repeats. It is a single-shot identification question rather than a graded judgement. Every call uses a fresh, empty context, so agreement or disagreement between conditions reflects the input alone.

Anonymisation. Candidate names, party affiliation and the debate date are removed from every input. Moderator turns are dropped entirely, leaving only candidate sentences in original chronological order. This is a partial defence at best. A transcript’s content—quoted lines, named policies, the other candidate’s real name spoken aloud mid-turn—was not scrubbed. Removing it would have taken out the communicative content the transcript condition exists to test. The contamination probe measures what leaks through this partial anonymisation rather than assuming it is complete. All 24 of 24 transcripts were identified regardless of the redaction.

Grounding check. A cited measure counts as consistent with a debate’s real data when the candidate it names as favoured actually has the higher within-debate standardised value for that measure. A margin of less than 0.15 standard deviations between the two candidates is scored as “similar” and excluded rather than forced to a side. This is an automatic, closed-vocabulary check, not a human judgement of whether the reasoning sounds plausible.

Appendix H Exploratory Same-Candidate Analysis

Appendix I 2024 Exclusion and Known Corpus Issues

Two 2024 debates exist in the corpus but are held out, because we could not verify who was speaking. Their official transcripts are paraphrased news summaries, sharing only 5.2% and 5.5% of their words with the audio. As a result, the recognised text was used as the transcript, and speakers were assigned by diarization [27]. That diarization was never checked against hand-labelled data. We have no error rate for it.

Four things prompted the exclusion. In the June 2024 debate one candidate is credited with 786 sentences and the other with 256—three to one, in a format with

Table H9 Where the same speaker lands when they won and when they lost. The discriminant was fitted *without* that speaker, then their debates projected in. Scores are relative to the opponent on the night.

Speaker	Debate	Outcome	Score
G. H. W. Bush	1988, first	won	-0.021
	1988, second	won	-0.066
	1992, first	lost	-0.224
	1992, second	lost	-0.203
	1992, third	lost	-0.223
	gap (won - lost) winning above losing		+0.173 6/6
Donald Trump	2016, first	won	+0.010
	2016, second	won	-0.018
	2016, third	won	-0.025
	2020, first	lost	-0.034
	2020, second	lost	+0.002
	gap (won - lost) winning above losing		+0.005 4/6

timed turns and muted microphones. In the September debate about 285 sentences sit in twelve or more leftover speaker clusters. Across the pair, 1,162 sentences are labelled *audience*, in two debates with no audience. Finally, candidates account for 43.9% of sentences in these two, against 76 to 86% everywhere else.

The confirmation is what happens on removal: across the remaining 24 debates, unattributed sentences drop to zero and audience sentences to 287. All of the unattributed material and 80.2% of the audience material came from those two debates. Including the pair changes several results. Noun rate flips sign, from -0.012 to +0.043. Sentiment roughly halves. The embedding axis rotates (cosine similarity 0.895 to the one reported here). Finally, 19 of the 50 most extreme sentences at one end come from this 9.5% of the data. These transcripts also contain no truncation marks and no filled pauses at all. As a result, interruption and self-correction are missing by construction.

Availability

The corpus is deposited with a permanent identifier. Released material covers the annotation layers, the per-sentence features and the checking scripts. Alignment scripts point at audio sourced elsewhere, which cannot be redistributed. A data statement, a column dictionary and the annotation codebook accompany the deposit.

Known problems

- 25 candidate sentences of eight words or more are exact duplicates within a debate, 17 of them in the first 1988 debate. One pair has the same text attributed to two different candidates, so speaker attribution is not verified anywhere in the corpus.
- 8 candidate rows share an identical time span—one group of nine sentences assigned a zero-length window.

- 238 candidate rows have no duration, almost all in one debate. That debate is also where every extraction error occurs.
- Acoustic extraction fails on 4.5% of candidate rows: 3.2% too short, 0.9% errors, split 2.8% against 16.5% across the two alignment treatments.
- No two candidate sentences overlap in time, so interruptions are not represented.
- Two rule-based categories duplicate the sentence-type column, and 13 codes never fire, both discussed in Appendix E.

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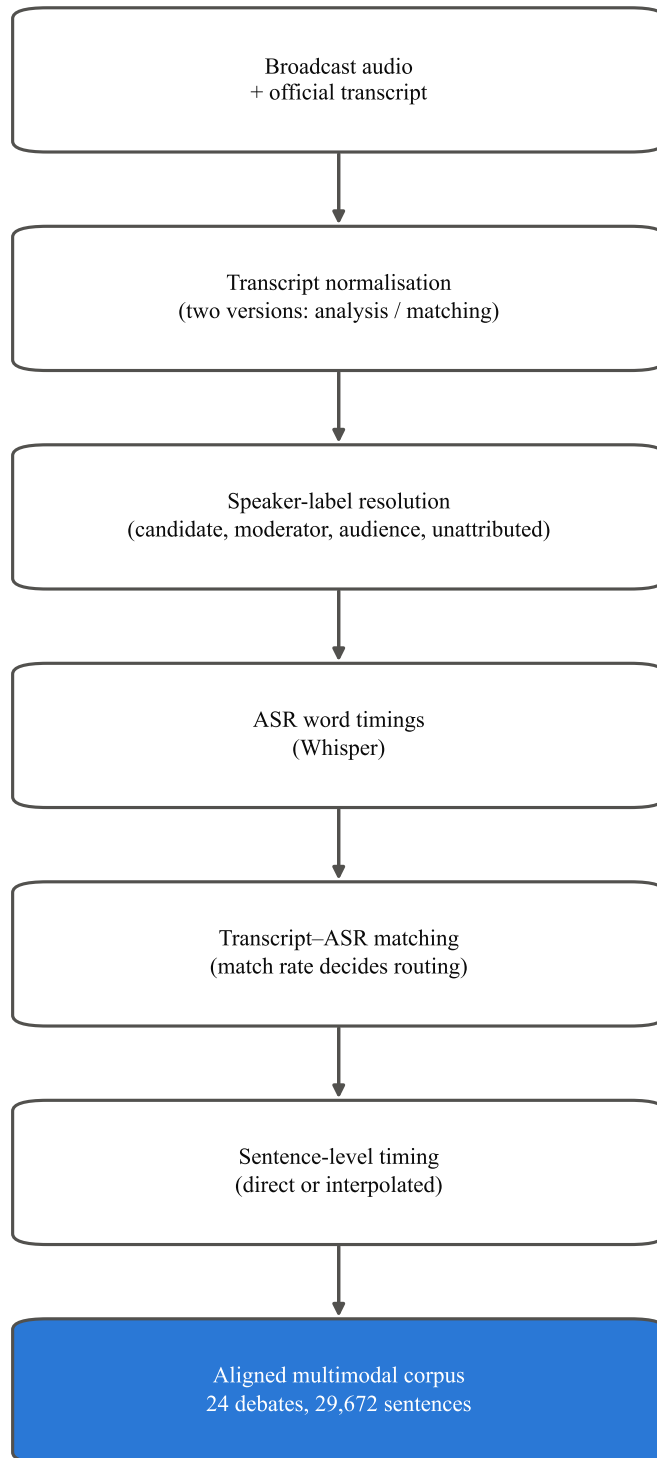


Fig. 1 Corpus construction pipeline. The match rate between the official transcript and recognised speech, computed before any analytical decision, determines whether a debate is routed to direct timing or interpolation (Table 3).

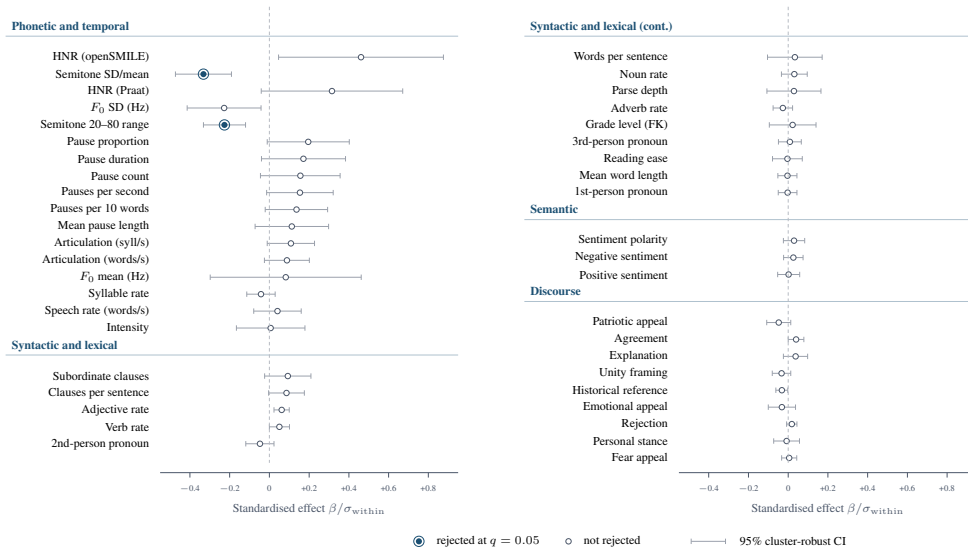


Fig. 2 Within-debate difference for all 43 measures, in standard deviations, with 95% confidence intervals that account for clustering by debate. Positive means higher among winning candidates. Filled markers are the two measures that survive correction for multiple comparisons. Note the scale: the biggest effect is 0.46 and most are under 0.05.

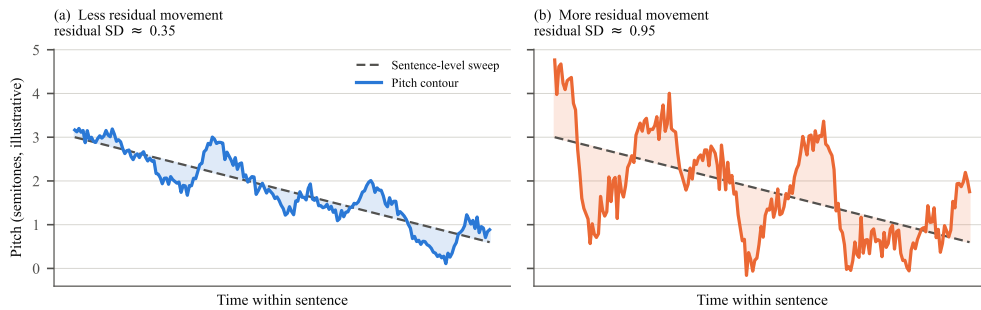


Fig. 3 What “residual pitch movement” means. Both panels share the same sentence-level sweep (dashed). The shaded area is the residual left over once that sweep is subtracted out. Synthetic illustration, not real corpus data.

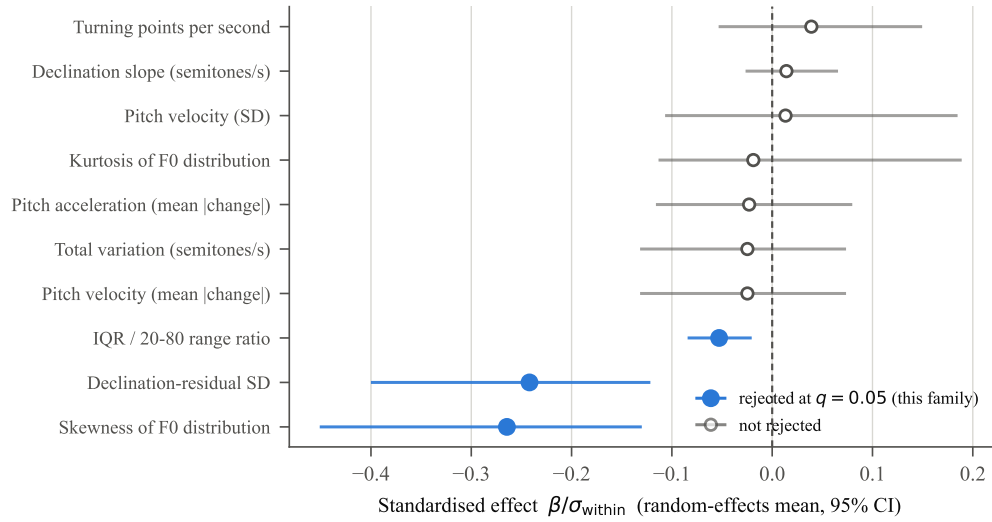


Fig. 4 Within-debate standardised effect (random-effects mean, 95% confidence interval) for the ten higher-order and dynamic pitch measures. Filled markers survive Benjamini–Hochberg correction within this family.

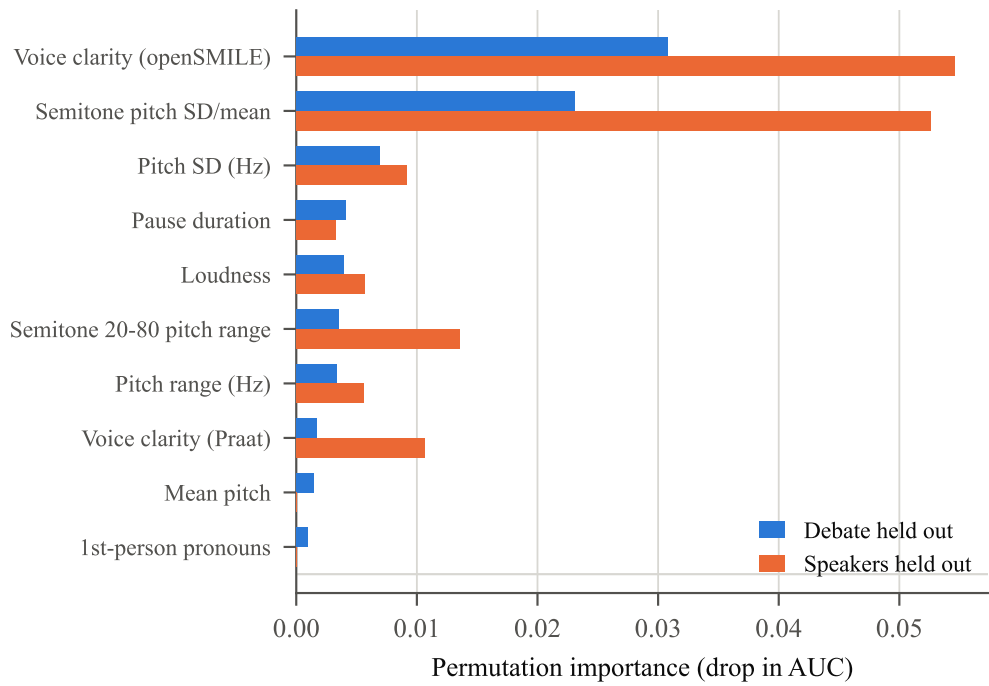


Fig. 5 Permutation importance of the top ten measures. The ranking is almost entirely acoustic, and it is led by the two measures shown in Section 5.3 to be tied to the speaker’s own voice.

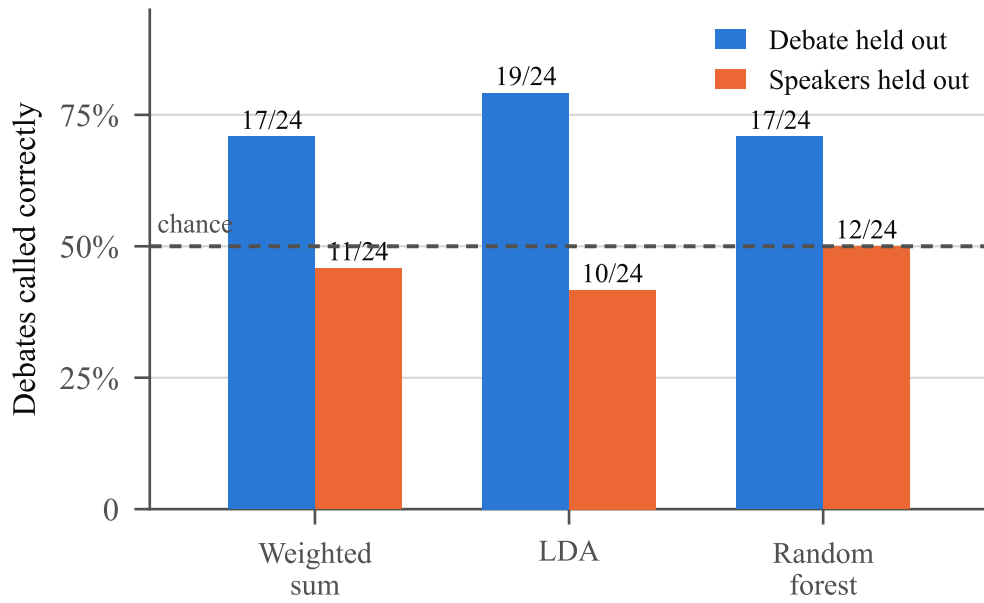


Fig. 6 The same three methods, evaluated two ways. Holding out the debate leaves both of its speakers in the training data; holding out the speakers removes them entirely. All three fall to chance under the stricter test.

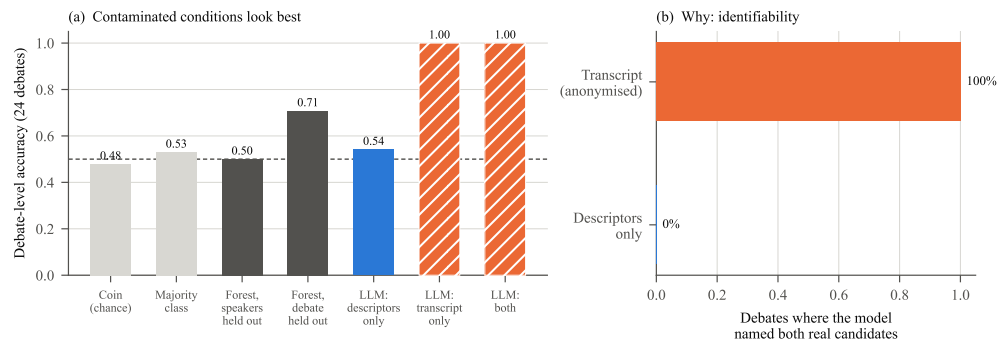


Fig. 7 (a) Debate-level accuracy of the language model, against the decision forest and the trivial baselines. The transcript condition (hatched) beats everything else in the paper. (b) shows why: the same model, asked only to identify the debate rather than predict its outcome, does so perfectly whenever it can see the transcript, and never when it can only see numbers.

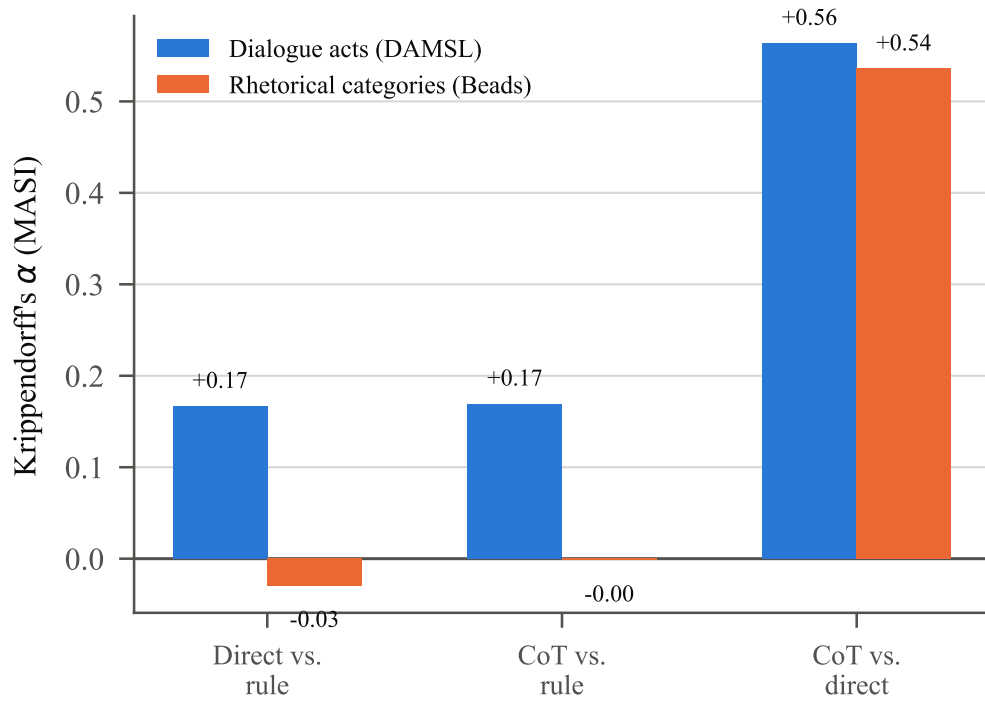


Fig. E1 Chain-of-thought prompting does not close the gap with the rule-based tagger. Both prompting styles land at the same chance-level agreement with the rules; what changes is how much the two LLM conditions agree with each other.

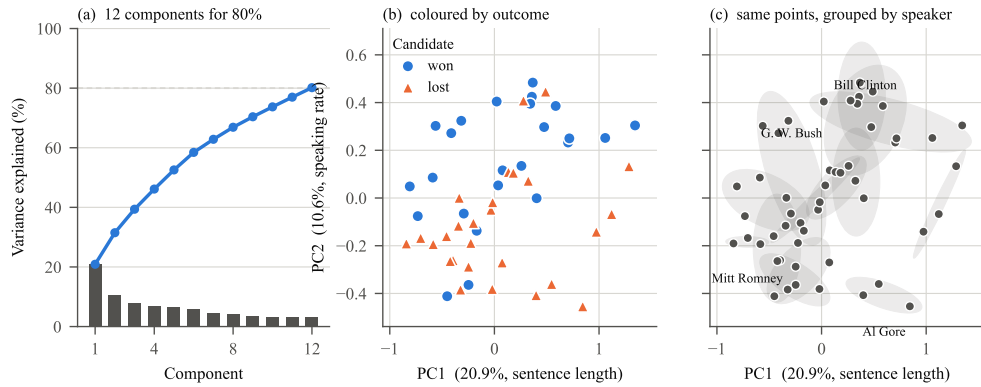


Fig. F2 Principal component analysis of the 34 standardised measures. (a) Twelve components are needed for 80% of the variance. (b) Each point is one candidate in one debate, coloured by outcome. (c) The same points, with an ellipse around each speaker's debates. The apparent separation by outcome in (b) coincides with the speaker grouping in (c).